New Mansoura University

Faculty of Engineering

Aeronautical & Aerospace Engineering Program

Graduation Project

Space Observatory Micro-Satellite & AOCS Testbed

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ABSTRACT

The Egyptian Gamma Observation Satellite (EGOS-1) is a space observatory micro-satellite designed and developed by Team Astro at New Mansoura University as part of the 2025–2026 graduation project for the Aeronautical and Aerospace Engineering Program. The project encompasses two parallel and complementary deliverables: the conceptual design and systems-level engineering of a gamma-ray observation micro-satellite, and the design, fabrication, and experimental validation of a ground-based Attitude and Orientation Control System (AOCS) evaluation testbed simulating the LEO magnetic and rotational environment.

EGOS-1 is a 20 × 20 × 20 cm micro-satellite designed for a sun-synchronous circular orbit at 550 km altitude. Its primary payload is a NaI(Tl) scintillation detector coupled to a BPW34 photodiode array for gamma-ray flux detection. Structural integrity was confirmed through static, thermal, and modal finite-element analyses, yielding a factor of safety of 7 under 10G launch loading, thermal compliance at a peak frame temperature of 411.51 K, and adequate dynamic stiffness across six vibration modes. Attitude control is achieved via a reaction wheel, dual custom magnetorquers, and a PSO-tuned PD controller executed on an ESP32 microcontroller, with real-time telemetry accessible through a Wi-Fi-hosted dashboard.

The AOCS testbed comprises a three-axis rectangular Helmholtz cage and a spherical aerostatic air bearing. The Helmholtz cage generates a uniform magnetic field in the LEO operational range (0.3–0.6 Gauss at 5 A), confirmed by Wolfram-based magnetic simulation and real-world testing. The air bearing, 3D-printed in PLA with 13 × 1.0 mm orifices at 6 bar supply pressure, was validated by independent SOLIDWORKS Flow Simulation and ANSYS Fluent 2025 R1 analyses, with both tools agreeing on a lifting force of 49.85 N to within 0.01%. Both the satellite Earth prototype and the testbed were successfully fabricated and demonstrated at New Mansoura University.

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1. Introduction

Space science has emerged as one of the most transformative fields driving scientific progress and technological development in the modern era. Over the past several decades, humanity's ability to deploy instruments and platforms beyond the confines of the Earth's atmosphere has fundamentally altered our understanding of the universe and has enabled a wide spectrum of applications that directly underpin daily life on Earth. From global telecommunications networks to precise navigation systems, from climate monitoring to disaster response, space-based infrastructure has become an essential element of the modern world's operational backbone.

The development and launch of artificial satellites represent arguably the single most consequential achievement of the space age. Since the launch of Sputnik 1 in 1957, satellites have evolved from simple radio beacons into sophisticated engineering platforms carrying instruments of extraordinary complexity and sensitivity. Today, constellations of satellites provide continuous coverage of the Earth's surface, enable real-time global communications, deliver precise positioning data to billions of users, and support scientific investigations ranging from Earth observation to deep-space astrophysics. The cumulative economic, social, and scientific impact of satellite technology is difficult to overstate BeiDou provide sub-meter positioning accuracy [14].

Artificial satellites serve a remarkable diversity of functions. In the domain of telecommunications, geostationary satellites relay voice, data, and broadcast signals across continents and oceans, forming the invisible backbone of the global internet and telephone infrastructure. In navigation, constellations such as the Global Positioning System (GPS), GLONASS, Galileo, and BeiDou provide sub-meter positioning accuracy that supports aviation, maritime navigation, precision agriculture, autonomous vehicles, and emergency services. In meteorology, polar-orbiting and geostationary weather satellites continuously monitor atmospheric conditions, enabling forecasts that save lives and protect economies. In Earth observation, multispectral and hyperspectral sensors map land use, monitor deforestation, track sea-ice extent, and assess agricultural yields, providing decision-makers with data critical for environmental stewardship and sustainable development [17].

Beyond these utilitarian applications, satellites have opened an entirely new window on the universe. Space-based observatories, freed from the blurring and absorptive effects of the Earth's atmosphere, observe the cosmos across the full electromagnetic spectrum, from gamma rays and X-rays through ultraviolet, visible, and infrared wavelengths to radio frequencies. Missions such as the Hubble Space Telescope, the Chandra X-ray Observatory, the Fermi Gamma-ray Space Telescope, and the James Webb Space Telescope have revolutionized astrophysics, cosmology, and planetary science, delivering data that challenge and refine our theoretical understanding of the universe's origin, structure, and evolution [20].

In recent years, the satellite industry has undergone a profound democratization driven by the rise of micro-satellites and CubeSats. Traditional large satellites, with masses measured in hundreds or thousands of kilograms and development costs in the hundreds of millions of dollars, have historically been accessible only to major space agencies and large commercial entities. Micro-satellites, defined as spacecraft with a mass between approximately 10 and 100 kilograms, and even smaller CubeSats based on standardized 10 × 10 × 10 cm unit (1U) form factors, have dramatically lowered the barriers to space access. Standardized launch vehicle interfaces, off-the-shelf electronic components, and streamlined development methodologies have enabled universities, research institutions, and technology start-ups to design, build, and launch space-qualified platforms on timescales of one to three years and at costs orders of magnitude lower than traditional satellite programs [15].

This democratization has had a catalytic effect on both scientific research and engineering education. Universities around the world now routinely develop and operate small satellites as educational platforms, training the next generation of aerospace engineers in systems integration, orbital mechanics, attitude control, communications, and mission operations. At the same time, micro-satellites are increasingly used for genuine scientific investigations, commercial Earth observation, technology demonstration, and in-orbit validation of new components and systems. The miniaturization of sensors, processors, and power systems has reached a point where a spacecraft the size of a shoebox can carry scientifically meaningful payloads and generate commercially valuable data.

Within the constellation of satellite subsystems, the Attitude and Orientation Control System (AOCS), sometimes referred to as the Attitude Determination and Control System (ADCS), occupies a position of particular criticality. The AOCS is responsible for sensing the satellite's current orientation in space, computing the required corrective maneuvers, and commanding actuators to reorient the spacecraft so that its instruments, antennas, and solar panels are pointed toward their intended targets with the required accuracy. For a space observatory, this pointing requirement is especially demanding: scientific instruments must be held precisely on a celestial target long enough to collect statistically meaningful data, and the pointing knowledge must be accurate enough to assign the collected data to the correct position on the sky.

A typical AOCS integrates several types of sensors to determine attitude. Magnetometers measure the local geomagnetic field vector, which can be compared to a model of the Earth's field to infer orientation. Gyroscopes, either mechanical or micro-electromechanical (MEMS), measure angular rates and provide high-bandwidth attitude rate information. Sun sensors detect the direction of the Sun relative to the spacecraft body frame. Star trackers, the most accurate of all attitude sensors, identify patterns of stars in their field of view and compare them to an onboard star catalog to determine inertial pointing with arc-second precision. On the actuator side, reaction wheels provide fine torque control about all three axes by exchanging angular momentum between the wheel and the spacecraft. Magnetorquers generate magnetic dipole moments that interact with the Earth's field to produce control torques, and are particularly useful for detumbling after launch and for desaturating reaction wheels [18].

The complexity of the AOCS, and the consequences of its failure for mission success, make thorough ground testing essential before a satellite is committed to orbit. Once in space, the satellite cannot be physically accessed for repair, and the cost of a failed mission is not merely the financial investment but the scientific opportunity that is irretrievably lost. Ground-based AOCS testbeds allow engineers to exercise sensors, actuators, and control algorithms in a controlled laboratory environment that simulates the key characteristics of the space environment: a frictionless rotational interface that mimics microgravity, and a controlled magnetic field environment that replicates the geomagnetic conditions the satellite will experience in orbit.

Egypt stands at a pivotal juncture in its development of national space capabilities. The Egyptian Space Agency (EgSA), established in 2018, has articulated a national space strategy that includes the development of indigenous satellite design and manufacturing capabilities. In this context, academic programs in aerospace engineering at Egyptian universities play a vital role in building the human capital that will realize these national aspirations. Projects that challenge students to design, analyze, and fabricate real space-qualified hardware contribute directly to the development of a domestic engineering talent pool capable of supporting Egypt's long-term space ambitions.

The Space Observatory Micro-Satellite and AOCS Evaluation Testbed project, designated EGOS-1, is designed in precisely this spirit. The project pursues two complementary objectives. The first is the conceptual design and analysis of a micro-satellite platform optimized for gamma-ray observation from Low Earth Orbit, with a scientific mission focused on the detection and characterization of gamma-ray flux from deep-space sources. The second is the design, analysis, and fabrication of a laboratory-based AOCS testbed capable of simulating the magnetic and rotational environment of LEO, enabling the rigorous validation of AOCS sensors, actuators, and control algorithms prior to any future launch campaign [21].

Gamma-ray astronomy represents one of the most scientifically rich and technically challenging domains of space-based observation. Gamma rays are the highest-energy photons in the electromagnetic spectrum, with energies typically exceeding 100 keV and often reaching into the GeV and TeV ranges. They are produced by the most energetic processes in the universe: the accretion of matter onto black holes and neutron stars, the annihilation of matter and antimatter, cosmic-ray interactions with interstellar medium, and the violent explosions of massive stars known as gamma-ray bursts. Because gamma rays are entirely absorbed by the Earth's atmosphere, they can only be detected by instruments deployed above the atmosphere, making space-based observatories an absolute requirement for gamma-ray astronomy.

The EGOS-1 payload employs a sodium iodide thallium-doped scintillation detector, a technology with a long and proven heritage in space-based gamma-ray detection. When a gamma-ray photon interacts with the NaI(Tl) crystal, it deposits its energy through a cascade of photoelectric, Compton scattering, and pair-production interactions, ultimately producing a burst of visible-wavelength scintillation light. This light is detected by an array of BPW34 high-sensitivity photodiodes mounted on the crystal's lower surface, whose output is amplified and digitized for onboard storage and subsequent downlink to the ground station.

The project's dual-component structure—satellite design and AOCS testbed development conducted in parallel and validated against a common set of mission requirements—reflects a systems engineering philosophy that is well established in the professional aerospace community. By developing the testbed in close coordination with the satellite design, the team ensures that the testbed's simulation capabilities are directly matched to the operational envelope of the AOCS that will fly on EGOS-1. Conversely, the testbed provides a concrete engineering product of immediate value for academic research, independent of whether and when the satellite itself is eventually launched.

The remainder of this report is organized as follows. Section 2 defines the problem statement that motivates the project. Section 3 states the project objectives. Section 4 reviews the relevant literature on microsatellite design, gamma-ray payload technology, and AOCS testbed approaches. Section 5 describes the engineering methodology adopted. Section 6 presents the technical approach, including design parameters, CAD models, simulation setups, and results for both the microsatellite and the AOCS testbed. Section 7 describes the deliverables and implementation plan. Section 8 identifies potential stakeholders. Sections 9 and 10 provide the project conclusion and future work, respectively.

1.1 Space Observatory Microsatellite

A space observatory micro-satellite is a compact, mass-constrained spacecraft specifically designed to observe and study astronomical objects and physical phenomena from a vantage point above the Earth's atmosphere. The absence of atmospheric absorption and distortion that afflicts ground-based observatories makes space-based platforms uniquely capable of detecting radiation across the full electromagnetic spectrum, including the gamma-ray, X-ray, and ultraviolet bands that are completely or partially blocked by the atmosphere at sea level.

Micro-satellites in this category typically have masses in the range of 10 to 100 kilograms, and their dimensions are constrained to fit within standardized launch vehicle dispensers. Despite these mass and volume constraints, modern micro-satellites can accommodate sophisticated scientific instruments, thanks to the continued miniaturization of electronics, detectors, and mechanical structures. Advanced system-on-chip processors provide the computational power needed for onboard data processing and compression. Low-power, high-sensitivity detectors enable meaningful science within tight power budgets. Compact attitude control actuators, including reaction wheels the size of a hockey puck, provide the pointing stability needed for scientific observation.

The EGOS-1 micro-satellite is designed to detect gamma-ray flux from astrophysical sources, contributing to the growing body of data from space-based gamma-ray observatories. Its orbit, subsystem architecture, structural design, and payload configuration are described in detail in Section 6 of this report. The satellite is designed with a mission lifetime of one to three years, depending on orbital decay and subsystem reliability, and is intended to serve as a demonstration platform for Egyptian space engineering capabilities as well as a scientific instrument contributing to international gamma-ray observation programs.

1.2 AOCS Testbed

The Attitude and Orientation Control System Testbed is a ground-based experimental platform engineered to simulate and evaluate the performance of a satellite's attitude control system in a controlled laboratory environment before the satellite is committed to space. In the context of the EGOS-1 mission, the AOCS must maintain precise pointing of the gamma-ray detector toward its intended observation targets, and it must do so reliably throughout the satellite's operational lifetime in the presence of environmental disturbances including aerodynamic drag, solar radiation pressure, gravity gradient torques, and geomagnetic perturbations.

The testbed enables the experimental validation of attitude sensors, actuators, and control algorithms by replicating two key aspects of the space environment. The Helmholtz cage, a system of three mutually orthogonal square coil pairs, generates a uniform and programmable magnetic field within the cage's central volume that replicates the Earth's geomagnetic field as experienced by a satellite in LEO. This allows magnetometers and magnetorquers to be tested and calibrated under realistic field conditions. The spherical air bearing provides a near-frictionless rotational interface on which a satellite mock-up can rotate freely about all three axes, simulating the near-zero-friction rotational environment of orbital free-fall.

Together, these two testbed components create a laboratory environment in which the full AOCS control loop—sensor measurement, state estimation, control law computation, and actuator command—can be exercised and validated before the satellite encounters the real space environment. The ability to detect and correct design errors at this stage, rather than after launch, is enormously cost-effective and is standard practice in professional satellite engineering programs worldwide.

2. Problem Statement

The exploration of space and the systematic observation of astronomical phenomena demand satellite systems of high precision, reliability, and operational robustness. Traditional large satellites, while capable of delivering exceptional scientific data, involve development costs measured in hundreds of millions of dollars and launch costs that are beyond the reach of universities and most national space agencies, particularly in the developing world. These financial barriers severely limit participation in space science and the development of indigenous satellite engineering expertise.

Micro-satellites represent a compelling cost-effective alternative that has the potential to democratize access to space-based observation. However, despite their lower cost, the technical challenges associated with micro-satellite development are not proportionally reduced. The accuracy and stability of satellite attitude control remain critical for mission success regardless of satellite scale, and the engineering difficulty of achieving precise, robust attitude control in a compact, power-constrained platform is substantial. Any systematic error or instability in the AOCS will result in degraded pointing accuracy, compromised data quality, and potentially mission failure.

In-orbit testing and debugging of AOCS failures is, in practice, extremely limited. Once a satellite is deployed, direct physical access is impossible, and the only means of diagnosing and correcting AOCS anomalies are telemetry analysis and software patch upload over the communication link. This underscores the critical importance of comprehensive ground testing before launch. However, the facilities needed for AOCS ground testing—Helmholtz cages, air bearing platforms, vibration tables, thermal-vacuum chambers—are expensive to procure and operate, and are typically found only at major space agencies and large aerospace companies.

This project addresses a specific gap: the absence, within Egyptian academic institutions, of a dedicated ground-based AOCS evaluation testbed that allows researchers and engineering students to design, test, and validate satellite attitude control systems under realistic simulated space conditions. By designing and fabricating such a testbed locally, using a combination of commercial off-the-shelf components and custom-designed mechanical parts, the project demonstrates that it is possible to create a functional, scientifically meaningful AOCS test facility at a cost accessible to university programs. Simultaneously, by designing the EGOS-1 microsatellite as the reference system for the testbed, the project produces a complete systems engineering case study that links mission requirements to testbed specifications.

3. Objectives

The main objectives of this project are:

To localize the fabrication of a high-efficiency, cost-effective Attitude and Orientation Control System (AOCS) testing apparatus, enabling reliable experimental validation of spacecraft attitude control subsystems. This involves the end-to-end design, manufacturing, and assembly of the testing platform using locally sourced components and domestic engineering expertise, thereby reducing dependence on imported systems. The apparatus will be capable of simulating realistic rotational dynamics and disturbance torques, providing a controlled environment in which AOCS algorithms and hardware configurations can be rigorously evaluated, calibrated, and refined prior to integration into the flight model.

To design a compact space observatory micro-satellite platform incorporating advanced sensors and payloads, with AI-enhanced data acquisition and interpretation for improved astronomical observation. The satellite platform will be engineered to meet the stringent mass, volume, and power constraints characteristic of the micro-satellite class, while accommodating a scientifically capable optical and sensing payload suite. The integration of artificial intelligence and machine learning techniques into the onboard data pipeline will enable autonomous feature extraction, anomaly detection, and prioritized downlink of high-value observational data, significantly enhancing the platform's scientific return within its operational constraints.

To provide affordable and accessible testing capabilities for universities and research institutions, supporting hands-on experimentation and academic training in satellite systems engineering. A central consideration in the design philosophy of the testing apparatus is its replicability and ease of use within academic settings that may have limited infrastructure or funding. By documenting the development process thoroughly and optimizing the system for low-cost production, this project aims to democratize access to spacecraft subsystem testing, enabling the next generation of aerospace engineers to gain practical, laboratory-grade experience with real-world satellite technologies.

CHAPTER 3: OBJECTIVES

To contribute to Egypt's advancement in space science and technology by designing and prototyping the nation's first space observatory micro-satellite, fostering local expertise and capability in aerospace engineering and scientific research. Beyond the immediate technical deliverables, this project is positioned as a foundational milestone in building a sustainable domestic space industry. Through the involvement of local engineers, academic institutions, and research bodies, the project seeks to cultivate a multidisciplinary knowledge base, stimulate innovation, and establish the technical heritage necessary for Egypt to undertake increasingly ambitious space missions in the future.

4. Literature Review

This chapter provides a comprehensive review of prior research and technological developments relevant to microsatellites, space observatory systems, and attitude determination and control systems (AOCS). A thorough examination of the existing literature is essential for understanding the current state of satellite technology, identifying proven design methods and approaches, and recognizing the limitations and challenges encountered in previous work. This review establishes a solid intellectual foundation for the proposed project and informs the design decisions made in the technical approach described in Chapter 6..

4.1 Space Observatory Microsatellite

4.1.1 Mission

Detection of gamma rays is one of the most effective methods for space observation, as these high-energy particles originate from deep space and carry valuable information about their sources . By analyzing the gamma-ray flux from specific regions, researchers can gain insights into the physical characteristics of these areas, enhancing our understanding of the universe. In line with this, the primary mission of the microsatellite will focus on gamma-ray detection [1]. To clearly reflect its purpose, the microsatellite has been named **The Egyptian Gamma Observation Satellite (EGOS-1)**.

4.1.2 Orbit Design

The selection of an appropriate orbit plays a critical role in the performance of gamma-ray instruments. Research indicates that near-equatorial, low-altitude circular orbits, typically ranging from 500 to 600 km, offer optimal conditions by minimizing background radiation [2]. In such Low Earth Orbits (LEO), the Earth's magnetic field provides effective shielding against cosmic rays and solar energetic particles, enhancing the accuracy and sensitivity of measurements. For the proposed space observatory microsatellite, this orbital regime achieves a favorable balance between scientific capability, radiation protection, operational stability, and cost efficiency, making it a practical and suitable choice for a small satellite platform.

4.1.3 Payload

Gamma-ray detector designs for small satellite missions typically employ arrays of scintillating crystals coupled with photosensors to achieve the combination of high detection efficiency, good energy resolution, and compact form factor required for space deployment. Arneodo et al. [3] present a reference detector architecture featuring a core of four scintillating crystal elements, each individually read out by a dedicated photosensor, housed within a compact aluminum enclosure that provides mechanical support and electromagnetic shielding. This architecture is adopted as the basis for the EGOS-1 payload design, with specific adaptations to accommodate the volume and mass constraints of the 20 × 20 × 20 cm satellite structure.

The EGOS-1 payload uses a sodium iodide thallium-doped (NaI(Tl)) scintillation crystal as its primary sensing element. NaI(Tl) is a well-characterized scintillator with a long heritage in space-based and laboratory gamma-ray detection, offering a favorable combination of high detection efficiency, adequate energy resolution, relatively low cost, and commercial availability in radiation-hardened grades. The scintillation light produced by gamma-ray interactions in the crystal is detected by BPW34 high-sensitivity photodiodes, whose output is amplified and digitized for onboard storage and downlink.

4.1.4 Sub-systems

A microsatellite, such as EGOS-1, consists of several key subsystems that work together to ensure mission success in Low Earth Orbit (LEO) [4]:

- **Onboard Computer (OBC):** Manages data from sensors and payload, executes ground commands, and coordinates all satellite operations.
- **Attitude and Orbit Control System (AOCS):** Maintains the satellite's orientation using sensors (magnetometers, gyroscopes, sun sensors) and actuators (reaction wheels, magnetorquers), enabling accurate pointing of instruments.
- **Communication (COMMS):** Handles data exchange between the satellite and ground stations, including telemetry, tracking, and command (TT&C).
- **Power Subsystem (EPS):** Supplies energy through solar panels and batteries, ensuring continuous operation and protecting sensitive electronics.

These subsystems are integrated to achieve reliable satellite performance and successful mission operations in LEO.

4.1.5 Frame & Housing

The structural frame of EGOS-1 serves as the primary mechanical load-bearing element of the satellite, providing mounting interfaces for all subsystems, defining the external envelope of the spacecraft, and protecting internal components from the mechanical loads experienced during launch [4]. The frame is designed as a compact cubic structure with external dimensions of 20 × 20 × 20 cm, fabricated from Aluminum Alloy 6061-T6 sheet of 1.5 mm thickness [4]. This alloy is widely used in satellite structural applications due to its high strength-to-weight ratio, excellent machinability, good corrosion resistance, and well-characterized behavior under the thermal cycling and radiation environments of LEO [4] .

The frame incorporates a 10 cm diameter circular access hatch on the top panel to facilitate assembly and integration of internal components, and dedicated mounting interfaces for two deployable solar panels of 16.5 × 10 cm each. The internal architecture follows a nested-stack configuration, in which subsystem boards are arranged in a layered vertical stack and fastened by mechanical standoffs, maximizing the use of available internal volume and simplifying the integration and testing procedure.

4.2 AOCS Testbed

4.2.1 Helmholtz Cage

Testing and calibrating the attitude determination and control sensors and actuators of a satellite before flight requires the ability to generate a controlled and uniform magnetic field in the laboratory that accurately replicates the Earth's geomagnetic field at the satellite's operational orbit. The Earth's magnetic field can be modeled as a magnetic dipole with a surface field strength of approximately 30,000 nT at the equator, decreasing with the cube of the geocentric distance and varying with magnetic latitude. For LEO microsatellites at altitudes of 400 to 600 km, the field strength at the satellite's location is typically in the range of 20,000 to 60,000 nT (0.2 to 0.6 Gauss), depending on orbital inclination and magnetic latitude.

For microsatellites in Low Earth Orbit (LEO), typical values can be obtained from the following equation [4] :

$$\mathbf{MF}(\mathbf{nT}) = \mathbf{30{,}000}^{*}\left(\mathbf{R}/(\mathbf{R}+\mathbf{H})\right)^{3*}\left[\mathbf{1}+\mathbf{3}^{*}\left(\sin(\varphi)\right)^{2}\right]^{0.5}$$

- Earth radius R = 6371 km
- Orbital altitude H = 400 – 600 km
- Magnetic latitude ϕ = 0° – 90°

The Helmholtz cage is the standard laboratory instrument used to generate this kind of controlled uniform magnetic field [5]. A classical Helmholtz cage consists of two identical circular coils, placed coaxially and separated by a distance equal to their radius. When equal currents flow through both coils in the same direction, the first-order spatial gradient of the magnetic field between the coil's cancels, producing a region of high field uniformity between and around the coils. A three-axis Helmholtz cage, consisting of three mutually orthogonal pairs of coils, can generate an arbitrary uniform magnetic field vector within its central working volume [5].

For the EGOS-1 AOCS testbed, square coil pairs are used instead of circular ones, because square coils produce a larger region of high field uniformity for a given coil aperture [5]. The field uniformity is optimized when the ratio of coil separation to coil side length is γ = 0.5445 [5], the value that eliminates the second-order spatial gradient term at the cage center. The three coil pairs are oriented in the XZ, XY, and YZ planes to enable independent control of the magnetic field vector in all three spatial directions.

A Helmholtz cage, made of three orthogonal coil pairs, can create uniform magnetic fields in its central volume. By adjusting the currents, it can replicate the Earth's field at different altitudes and latitudes, allowing calibration of magnetometers, testing of magnetic actuators, and validation of AOCS algorithms before LEO deployment [5].

4.2.2 Spherical Air Bearing

The spherical air bearing is the second principal component of the AOCS testbed. Its function is to provide a mechanical interface on which a satellite mock-up can rotate freely about all three axes with minimal friction, simulating the near-frictionless rotational freedom that a satellite experiences in the microgravity environment of orbit. Without such a frictionless support, ground-based tests of AOCS algorithms would be dominated by friction torques from the bearing that are many orders of magnitude larger than the disturbance torques the AOCS must reject in orbit, making the test results misleading and non-representative [6].

The spherical air bearing operates on the aerostatic principle: compressed air is fed through a network of orifices distributed over the inner surface of a hemispherical bowl, forming a thin film of pressurized air between the bowl and a spherical ball on which the satellite mock-up is mounted. This air film supports the weight of the satellite mock-up and eliminates solid-to-solid contact, reducing the friction coefficient at the bearing interface to values in the range of 10^{-4} to 10^{-5}, comparable to the disturbance torque environment in LEO. The design of the air bearing, including the selection of orifice diameter, number, and distribution, the plenum volume, and the supply pressure, must be carefully optimized to achieve both adequate lifting force and a sufficiently uniform air film pressure distribution [6].

5. Methodology

This project adopts a comprehensive and structured engineering methodology for the design, development, and validation of the Egyptian Gamma Observation Satellite (EGOS-1), along with its Attitude Determination and Control System (AOCS) testbed. The methodology combines theoretical analysis, system-level design, and experimental validation to ensure that the proposed system achieves its mission objectives with high reliability and accuracy.

The methodology is centered around two main components: the microsatellite system and the AOCS testbed. These components are developed in parallel and are closely interconnected throughout the project lifecycle.

The first component focuses on the design and development of the microsatellite. This begins with defining the mission objective, which is the detection of gamma rays in Low Earth Orbit (LEO). Based on this objective, a set of system requirements is established, including constraints related to power consumption, communication capabilities, and attitude control accuracy. The microsatellite is then structured into its primary subsystems: the Onboard Computer (OBC), which manages data processing and system operations; the AOCS, which controls the satellite's orientation; the Communication subsystem, which enables data transmission with ground stations; and the Power subsystem, which ensures continuous energy supply. Each subsystem is carefully designed and analyzed to ensure compatibility and efficient integration within the overall system.

The second component of the methodology involves the development of the AOCS testbed. Since the AOCS plays a critical role in maintaining the correct orientation of the satellite, it must be thoroughly validated before deployment. For this purpose, a laboratory-based testbed is designed to simulate the space environment. A key element of this setup is the Helmholtz cage, which is used to generate a controlled and uniform magnetic field that replicates the Earth's geomagnetic field. This allows accurate testing and calibration of AOCS sensors, such as magnetometers, and actuators, such as magnetorquers. The testbed provides a controlled environment where the performance of attitude control algorithms can be evaluated under repeatable conditions.

In parallel with both components, mathematical modeling and simulation are conducted. These simulations are used to analyze the behavior of the microsatellite and its AOCS under different operational scenarios, including variations in orbital conditions and external disturbances. The simulation results are then compared with experimental data obtained from the AOCS testbed to ensure consistency and validate system performance.

Finally, the microsatellite subsystems and the AOCS testbed are integrated and tested as a complete system. This stage ensures that all components function together as intended and that the system meets the predefined mission requirements. This combined approach of design, simulation, and experimental validation significantly reduces risks and enhances the reliability of the proposed system.

5.1 Design Framework

The design framework represents the practical implementation of the methodology and provides a clear and organized structure for the project workflow. It is based on a top-down approach and is divided into two main parallel tracks: the microsatellite development track and the AOCS testbed development track. These tracks are interconnected and converge during the integration and validation phase. Shown in Figure 1 is the design steps for both systems:

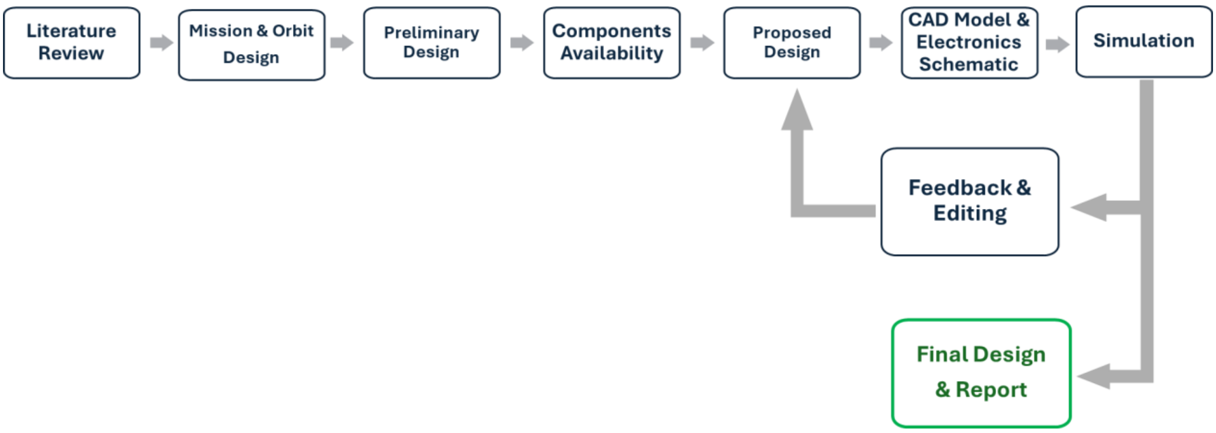


Figure 1 Workflow diagram for the technical design of the project.

5.1.1 Microsatellite Development Track

This track focuses on the complete design and development of the EGOS-1 microsatellite. It begins with mission definition and requirement analysis, where the objectives of gamma-ray detection are translated into technical specifications. The system architecture is then developed by dividing the satellite into its main subsystems (OBC, AOCS, Communication, and Power). Each subsystem is designed and analyzed individually, followed by system-level integration. Modeling and simulation are also performed within this track to evaluate system performance and ensure that the design meets mission requirements.

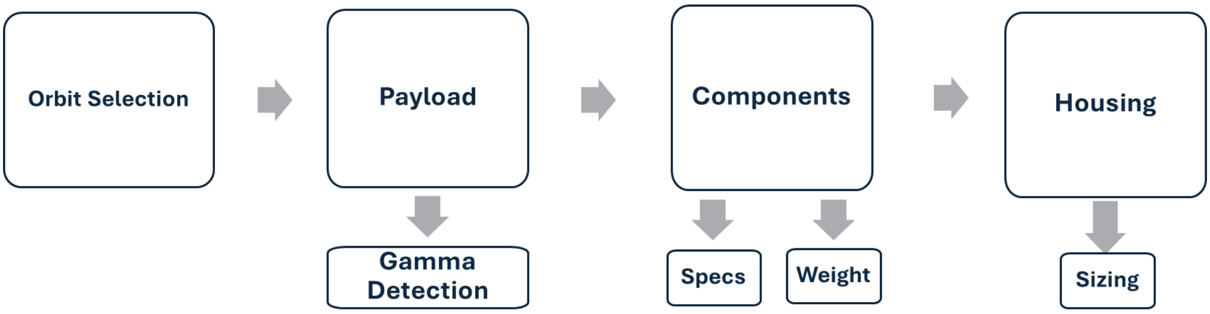


Figure 2 Workflow diagram for the design process of EGOS-1.

5.1.2 AOCS Testbed Development Track

This track is dedicated to the experimental validation of the attitude control system. It involves designing and implementing the AOCS testbed, including the Helmholtz cage for magnetic field simulation. The testbed is used to calibrate sensors, test actuators, and validate control algorithms. This experimental approach ensures that the AOCS performs accurately in a controlled environment before being applied to the actual microsatellite system.

5.1.3 Integration and Validation

In this final stage, both tracks are combined. The results obtained from simulations and experimental testing are compared and analyzed to evaluate overall system performance. Any discrepancies are addressed through design refinement and optimization. This stage ensures that the microsatellite and its AOCS meet all mission requirements and are ready for deployment in Low Earth Orbit.

5.2 Gantt Chart

The project is planned across two academic semesters. The first semester covers the design, modelling, and simulation phase, while the second semester focuses on implementation and hardware fabrication.

Figure 3 presents the Gantt chart for the first semester activities, covering the design, modelling, and simulation phase of the project, with tasks distributed across all team members. While Figure 4 presents the Gantt chart for the second semester, covering the implementation phase including hardware procurement, fabrication, assembly, and testing activities.

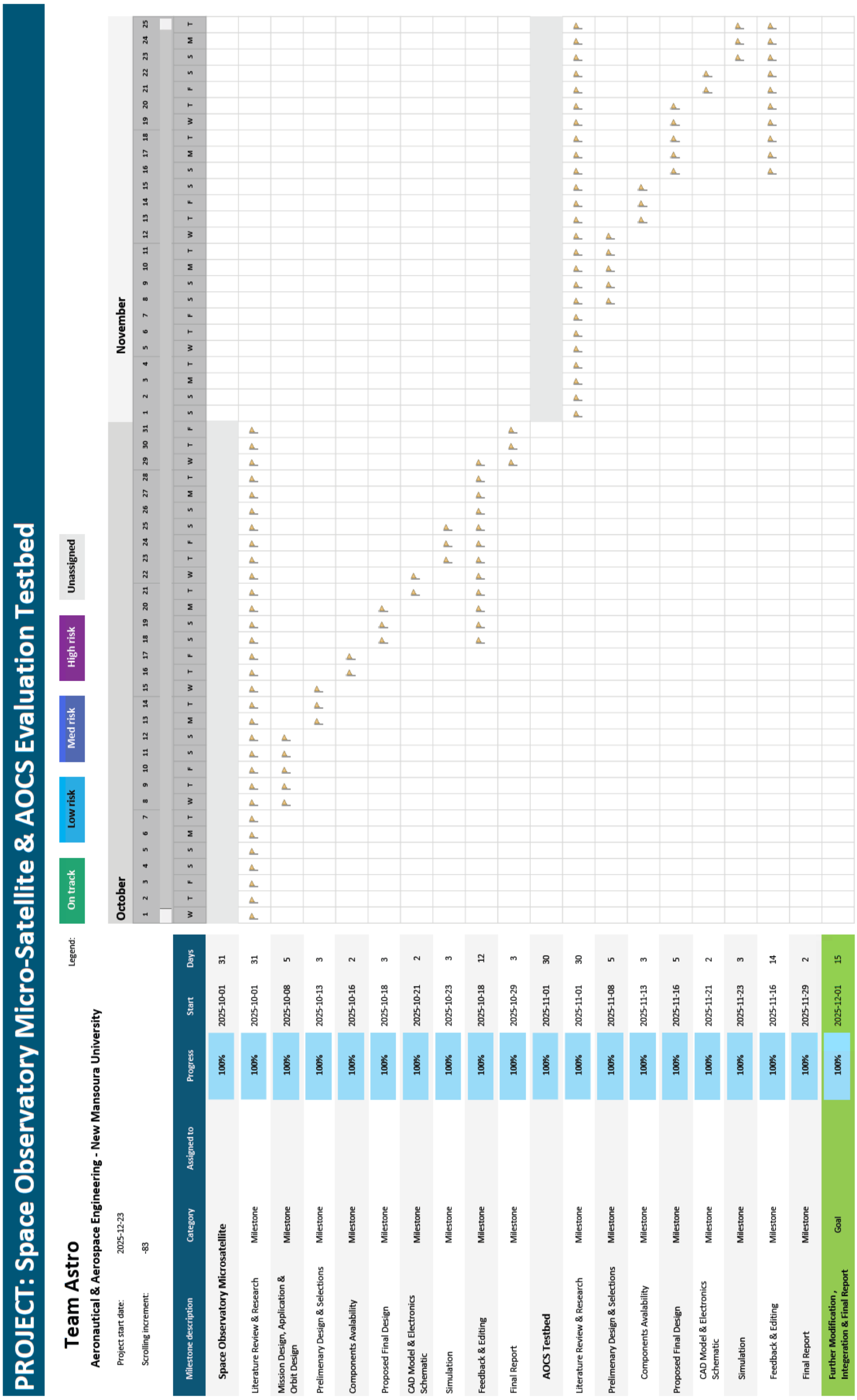


Figure 3 First semester Gantt Chart (Designing, Modelling & Simulation Phase)

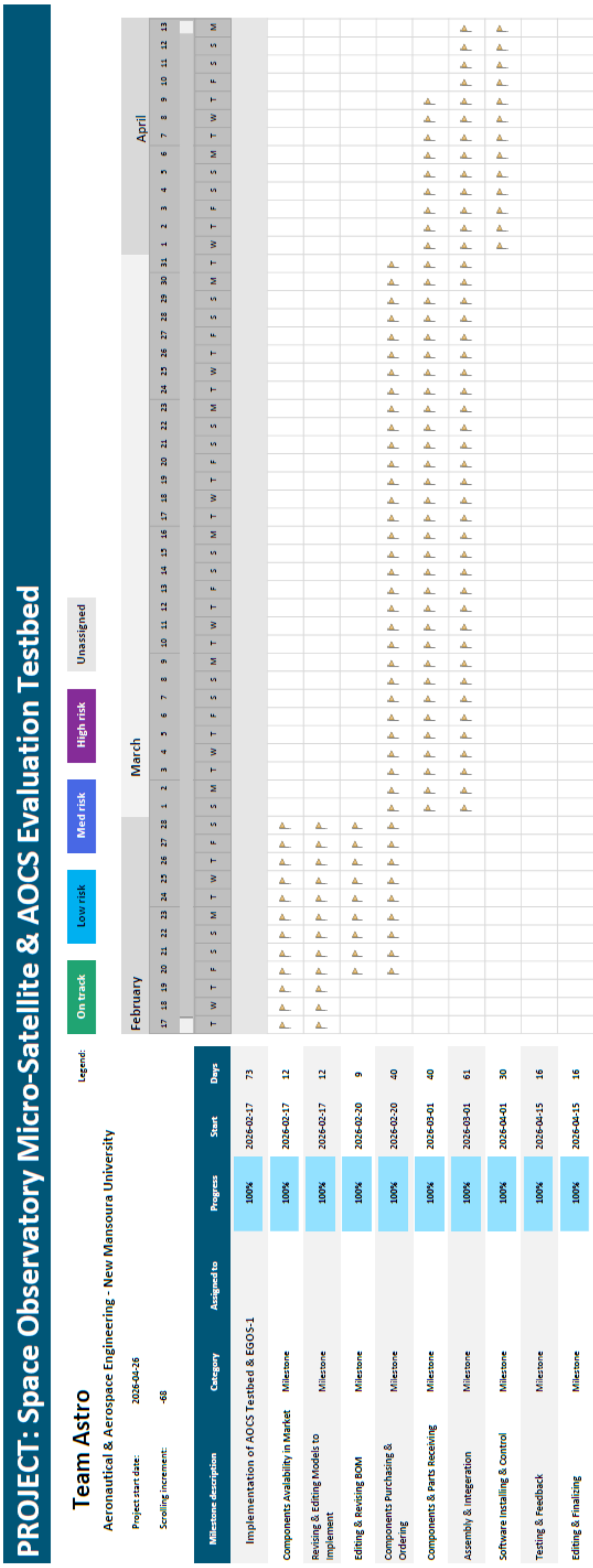


Figure 4 Second semester Gantt Chart (Implementation Phase)

6. Technical Approach

This chapter presents the technical methodology adopted for the design, modelling, and simulation of **The Space Observatory Microsatellite** and **The Attitude And Orientation Control Subsystem (AOCS) Testbed**. The approach follows a systems engineering methodology, integrating structural design, subsystem configuration, attitude dynamics modelling, and numerical simulation. The main objective is to ensure that the proposed microsatellite meets the mission requirements, particularly in terms of stability, pointing accuracy, and operational feasibility for space observation applications. In addition to the complete successful operation of the AOCS testbed.

6.1 Space Observatory Microsatellite

6.1.1 Design Parameters

The design parameters define the fundamental characteristics and operational constraints of the microsatellite. These parameters are carefully selected to achieve optimal system performance while ensuring feasibility within the limitations of microsatellite platforms.

The satellite mass and dimensions are minimized to comply with launch requirements and to improve overall system efficiency. The power subsystem is designed based on solar energy generation and onboard battery storage, ensuring sufficient energy availability for continuous subsystem operation.

The payload is designed to achieve high sensitivity and accuracy within constrained resources, making it suitable for space observation missions. Communication bandwidth is optimized to enable efficient data transmission between the satellite and ground stations.

Additional parameters include thermal operating limits, structural integrity, and mission lifetime, which is typically expected to range from one to three years depending on environmental conditions and system reliability.

6.1.2 Design Environment

The Space Observatory Microsatellite is designed to operate in the Low Earth Orbit (LEO) environment, which is characterized by harsh and highly variable space conditions. These include vacuum, radiation exposure, microgravity, and significant thermal cycling due to alternating periods of sunlight and eclipse. Such environmental factors have a direct influence on the design of the satellite subsystems, particularly the scientific payload.

Within this environment, the payload is specifically designed for gamma-ray detection using a sodium iodide thallium-doped (NaI(Tl)) scintillation crystal. The crystal functions as the primary sensing element, where incident gamma radiation interacts with the material and is converted into visible light through the scintillation process.

To operate effectively under space conditions, the NaI(Tl) crystal is enclosed within an aluminum casing, providing mechanical protection and environmental shielding. The optical signals generated by the crystal are detected using multiple BPW34 high-sensitivity photodiodes mounted on the lower surface of the crystal. These photodiodes act as transducers, converting light signals into electrical signals for further processing.

The resulting electrical output is then passed through an amplification stage to enhance signal strength and improve measurement readability prior to data acquisition and storage within the onboard system. This signal conditioning process is essential to ensure accurate data interpretation under the noisy and high-radiation conditions of the space environment.

Overall, the design of the payload is closely driven by the constraints of the LEO environment, where radiation tolerance, thermal stability, and structural robustness are critical factors affecting system performance and reliability.

The following Table 1 illustrates the orbit parameters values for EGOS-1:

Table 1 EGOS-1 orbit parameters

Parameter	Symbol	Value	Unit
Orbital Altitude	h	550	km
Eccentricity	e	0	-
Inclination	i	98.1	degrees
Right Ascension of Ascending Node	Ω	70	degrees
Orbital Period	T_{orbit}	95.7	minutes
Orbital Velocity	S_{orbit}	7.6	km/s

The orbital parameters mentioned in Table 1 define the characteristics and motion of the satellite in its orbit around Earth. The orbital altitude (h) represents the height of the satellite above the Earth's surface, which in this case is 550 km. The eccentricity (e) describes the shape of the orbit; a value of zero indicates a perfectly circular orbit. The inclination (i) is the angle between the satellite's orbital plane and the Earth's equatorial plane, determining the ground coverage of the satellite and here measured as 98.1°. The right ascension of the ascending node (Ω) defines the orientation of the orbit in space and represents the angle in the equatorial plane from the reference direction (vernal equinox) to the ascending node, where the satellite crosses the equator moving northward, with a value of 70°. The orbital period (T orbit) is the time required for the satellite to complete one full revolution around the Earth, which is approximately 95.7 minutes. Finally, the orbital velocity (S orbit) represents the speed of the satellite as it travels along its orbit, which is approximately 7.6 km/s for this configuration [4].

Using the SatCatalog design tool, the satellite's ground track was generated to analyze its orbital path illustrated in Figure 5. Additionally, the eclipse duration was calculated illustrated in Figure 6, representing the period during which the satellite operates without solar power and relies entirely on its batteries.

This parameter is critical, as it directly impacts battery sizing and ensures reliable operation during periods of solar unavailability.

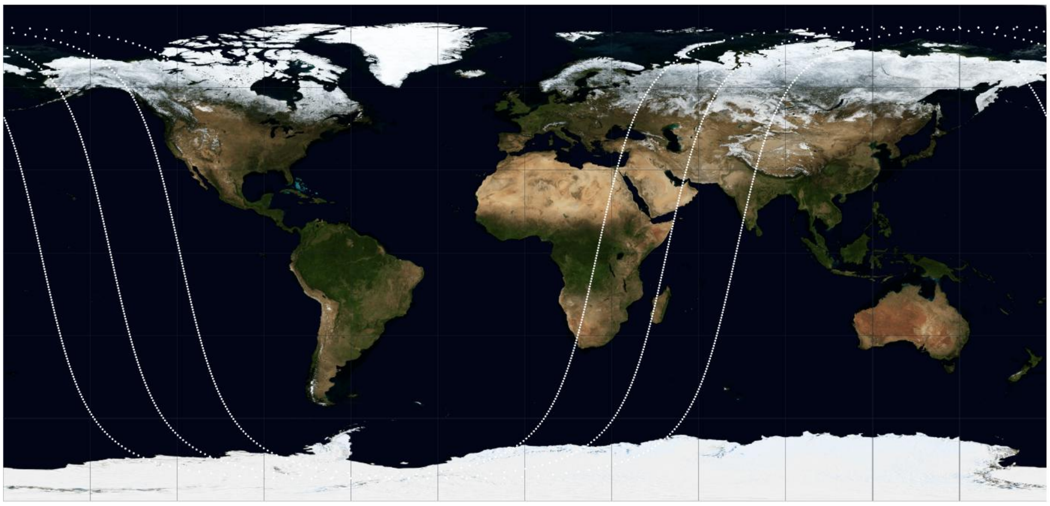
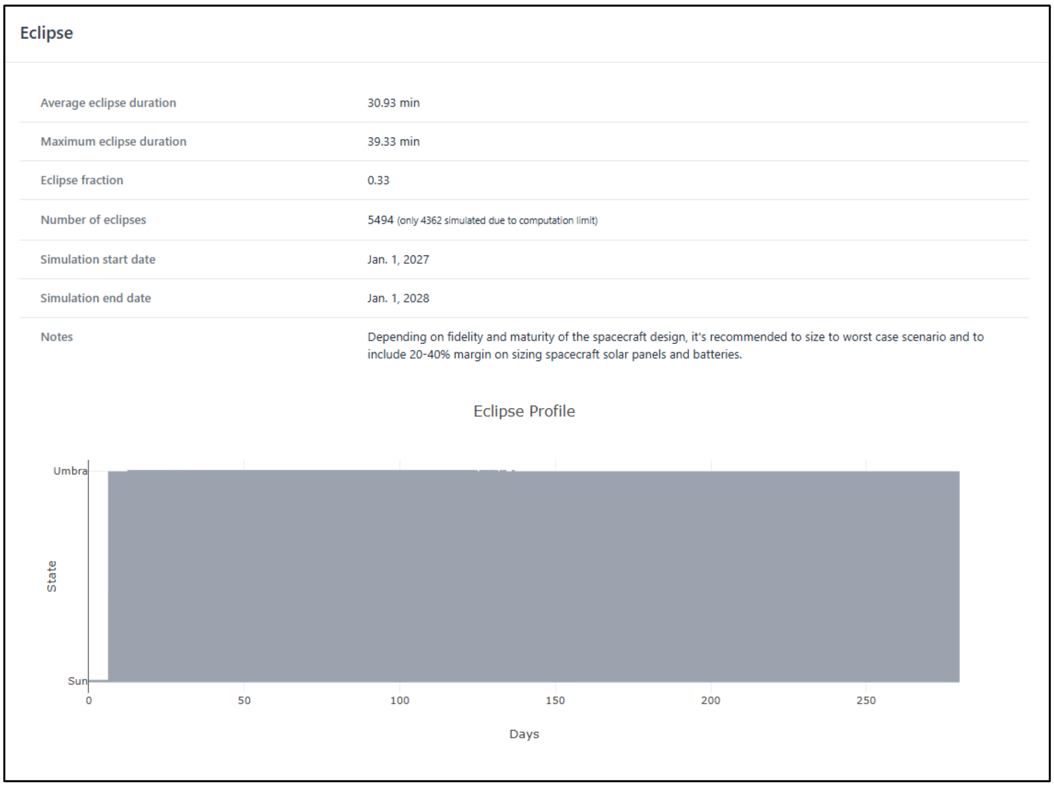


Figure 5 EGOS-1 ground track illustrating the satellite passing over Egypt for the downlink process.



Eclipse

Average eclipse duration	30.93 min
Maximum eclipse duration	39.33 min
Eclipse fraction	0.33
Number of eclipses	5494 (only 4362 simulated due to computation limit)
Simulation start date	Jan. 1, 2027
Simulation end date	Jan. 1, 2028
Notes	Depending on fidelity and maturity of the spacecraft design, it's recommended to size to worst case scenario and to include 20-40% margin on sizing spacecraft solar panels and batteries.

Figure 6 Eclipse simulation for EGOS-1 in orbit using SatCatalog. The maximum eclipse duration is around 40 minutes.

6.1.3 Payload Design

The payload design consists of a compact, robust radiation sensing module centered around a thallium-activated sodium iodide (NaI(Tl)) scintillation crystal, selected for its high light yield and proven efficiency in gamma-ray detection. The crystal is securely encapsulated within a cylindrical aluminum housing, which provides mechanical protection, electromagnetic shielding, and effective thermal stability while maintaining a lightweight profile suitable for UAV integration. The lower section of the assembly incorporates an array of BPW34 photodiodes, strategically arranged to maximize photon collection from scintillation events and ensure uniform sensitivity across the crystal volume. This configuration enables efficient conversion of incident radiation into measurable electrical signals with minimal loss. The modular structure, as illustrated in Figure 7 through the 3D, front, and bottom views, facilitates ease of assembly, maintenance, and potential scalability, while ensuring reliable operation under varying environmental and flight conditions.

Figure 7 The payload sensor consisting of a NaI(Tl) crystal incapsulated in an aluminum case (transparent in the figure) and connected to the BPW34 light sensors at the bottom. Showing a 3D, Front and Bottom View.

6.1.4 CAD Model & Schematic

The CAD model of the Space Observatory Microsatellite was developed to provide a comprehensive three-dimensional representation of the system architecture, including the structural frame, payload integration, and subsystem layout. The model serves as a critical design tool for verifying mechanical compatibility, spatial arrangement, and overall system integration prior to fabrication.

The structural configuration for the EGOS-1 frame defines the primary mechanical support of the satellite. The frame dimensions are 20 × 20 × 20 cm illustrated in

Figure *8*, forming a compact cubic structure suitable for microsatellite applications. The structure's material is Aluminum Alloy 6061-T6 with a thickness of 1.5 mm, selected for its high strength-to-weight ratio, mechanical durability, and suitability for space environments.

The design includes a 10 cm diameter access hatch located on the top panel to facilitate assembly, inspection, and integration of internal components. Additionally, dedicated mounting interfaces are incorporated to accommodate two deployable solar panels with dimensions of 16.5 × 10 cm each, ensuring efficient energy generation while maintaining compact stowage during launch.

The internal architecture follows a nested stack configuration, where subsystems are arranged in layered form to optimize space utilization and maintain a structured integration layout. The assembly is realized using mechanical fasteners, allowing modular construction and ease of maintenance or modification.

Complementary to the CAD model, schematic diagrams are developed to represent the functional and electrical architecture of the microsatellite. These schematics define the interconnection between subsystems, including power distribution networks, communication links, payload interfaces, and onboard data handling systems.

Overall, the CAD model and schematic representation provide a complete engineering framework for system validation, ensuring proper integration between mechanical design and functional requirements, and supporting subsequent simulation and implementation phases.

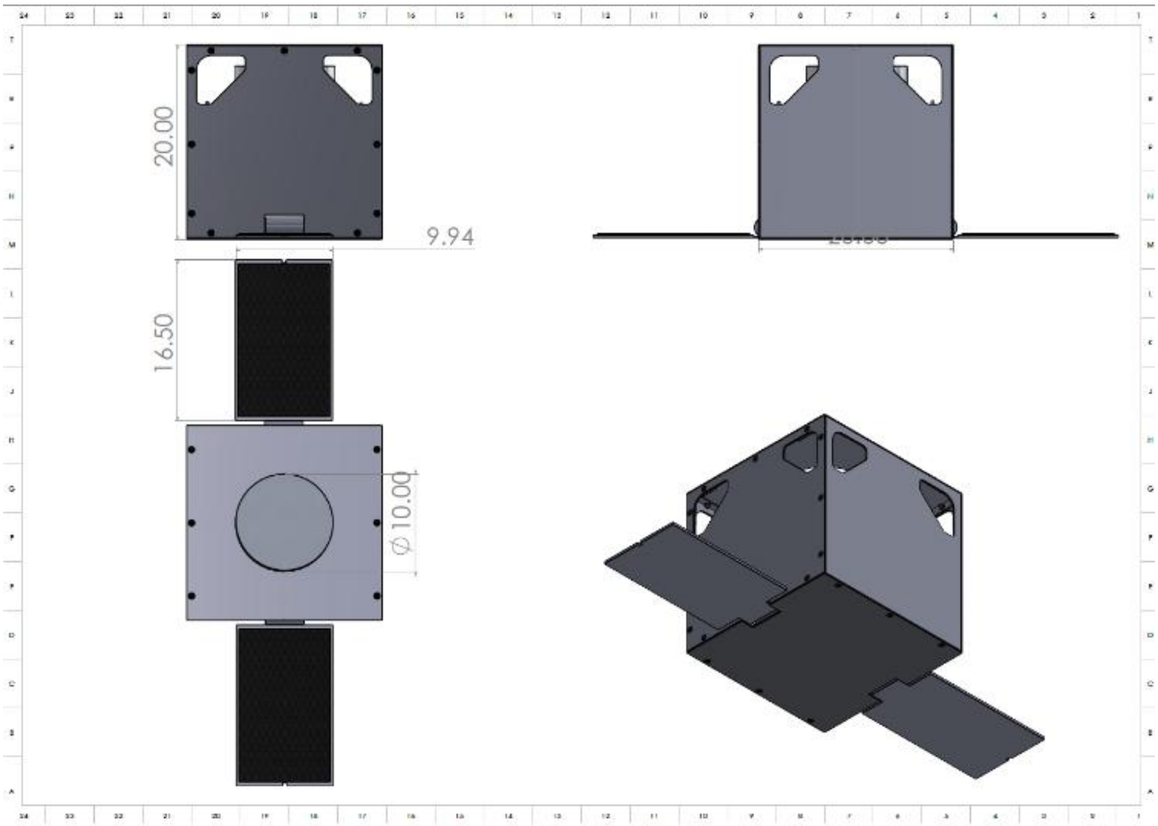


Figure 8 EGOS-1 frame dimensions in centimeters.

The satellite subsystems are designed to be fully interconnected and integrated with the onboard computer, which serves as the central processing unit of the system. This central unit is responsible for processing and analyzing incoming data, executing control commands for satellite operation, and managing communication between the satellite and the ground station, including both telemetry transmission and command reception.

To ensure a clear and structured system architecture, a complete electronic schematic was developed using Proteus 9 software illustrated in Figure 9. This schematic includes all satellite subsystems and defines the electrical and functional connections between them, providing a detailed representation of the overall system integration.

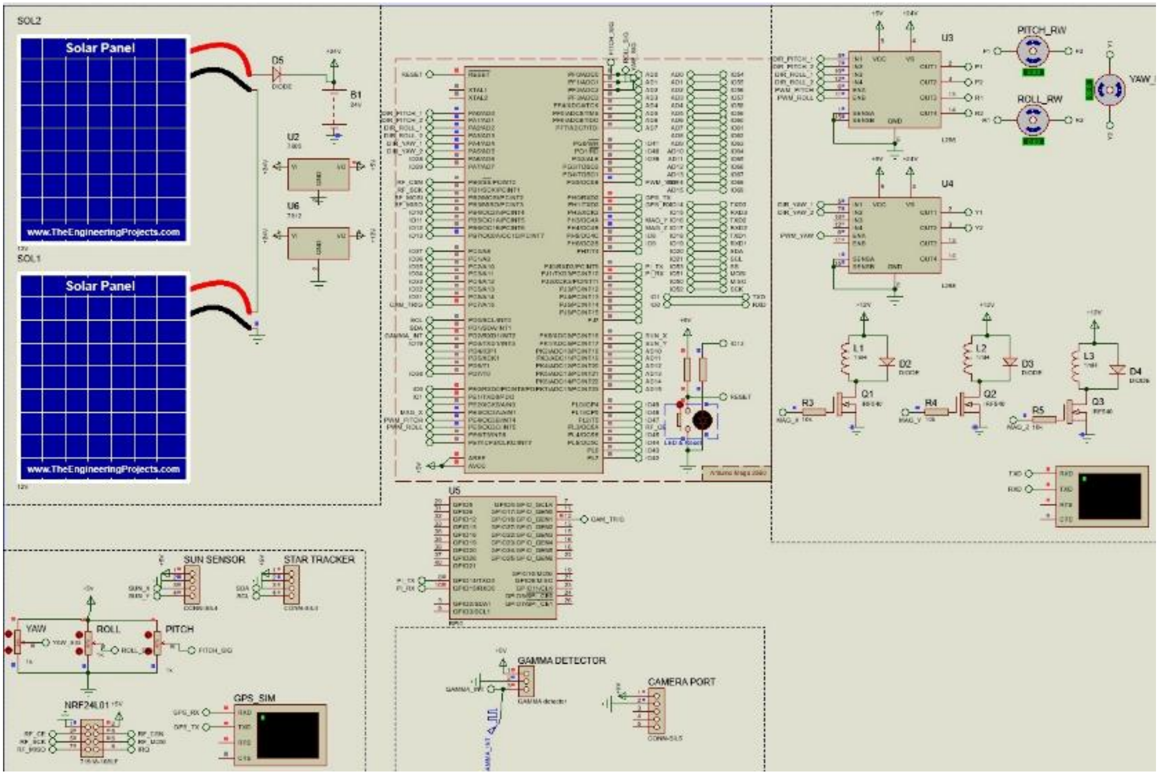


Figure 9 EGOS-1 full electrical schematic on Proteus 9 software.

For the nested stack structure, the schematic was converted into a 3D PCB model to visualize the placement of components illustrated in Figure 10, interconnections, and embedded systems within the EGOS-1 microsatellite Earth prototype, ensuring proper integration and spatial compatibility.

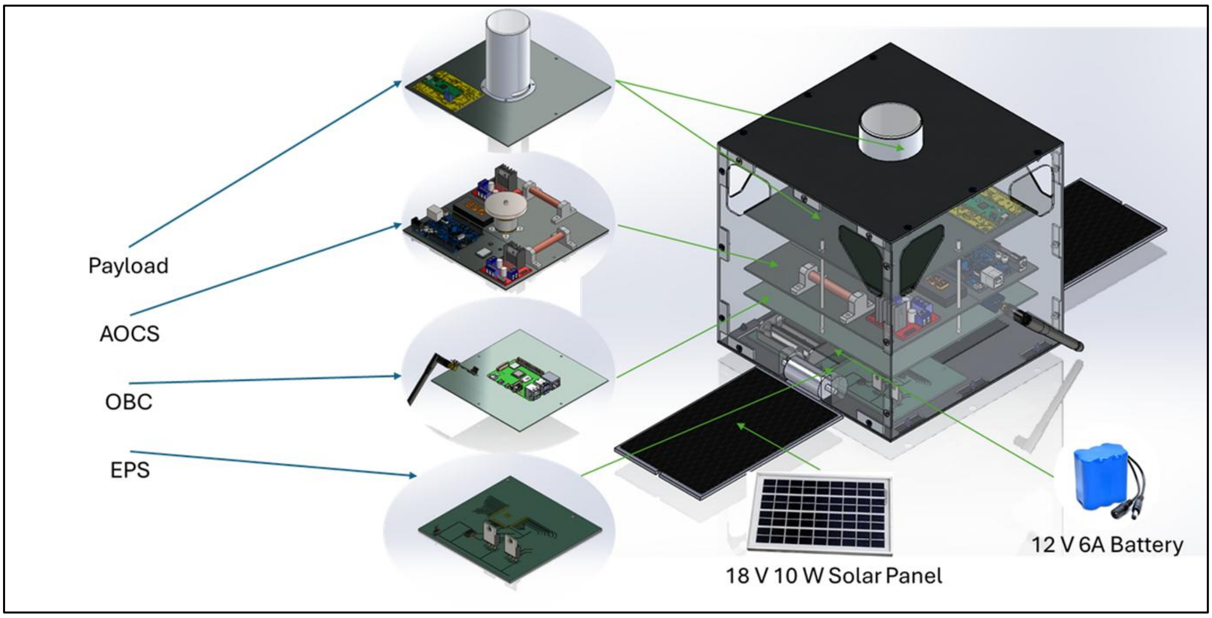


Figure 10 Exploded view of EGOS-1 illustrating all sub-systems and components in a nest stack structure with two plates in a transparent view.

6.1.5 Simulations Setup

To ensure that EGOS-1 will not fail in mission thorough different scenarios especially during launch, deployment and operation, several simulations were conducted to ensure the safety of this microsatellite. Static, thermal and modal simulations were conducted on the frame.

Structural finite element analysis was performed in SolidWorks Simulation to evaluate the mechanical integrity of the EGOS-1 frame under launch loading conditions. The launch phase represents the most mechanically demanding event in the satellite's lifecycle, during which the structure must withstand quasi-static and dynamic loads significantly exceeding normal operational levels. The simulation applies boundary conditions representative of the loads transmitted through the launch vehicle dispenser interface, using material properties for Aluminum Alloy 6061-T6.

6.1.5.1 Static Simulation

During the launching phase, the satellite undergoes high G forces that can reach up to 10 G's. Therefore, it's really important to ensure that the frame will not deform or fail during launching [4],[7],[8],[9],[10],[11].

Static simulation has been conducted on SolidWorks Simulation to acquire the deflection, maximum stress and the factor of safety of the frame, in addition to studying the hotspots on the frame. Boundary conditions were applied as loads and fixtures, an external load of 10G was added, and the bottom plate of the frame was fixed to simulate its attachment to the rocket. Proceeding with the simulation steps, the model was meshed with the elements having an aspect ratio ranged between 1.020 to 11.036 as shown in Figure 11, this big difference in the data range is due to limited computational power.

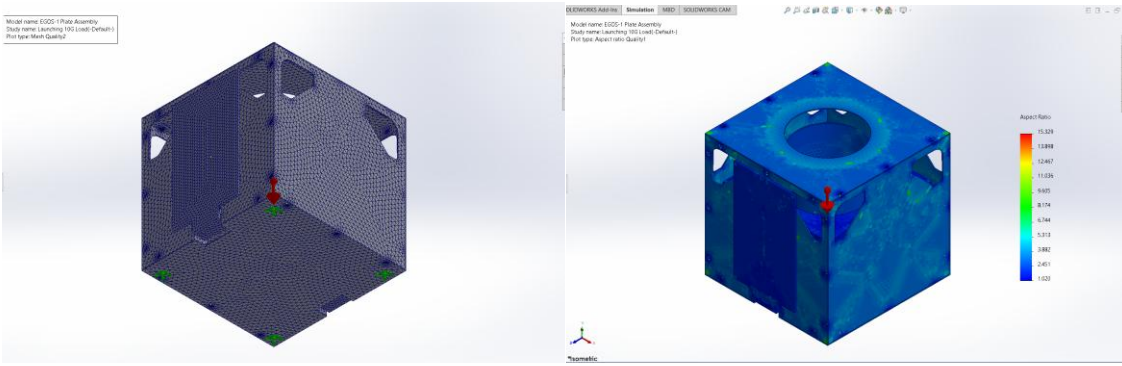


Figure 11 Mesh quality plot and boundary conditions showing the red arrow as the 10G external load and the green arrows as the fixture on the bottom plate in the left image. Mesh quality plot in aspect ratio in the right image.

6.1.5.2 Thermal Analysis

After the deployment of the satellite in orbit, it's exposed to the sun from one side and the other side is exposed to darkness and extreme cold weather, this variation in heat and temperature should be simulated to ensure the stability of the temperature and if it's the right value for the components to operate in [12].

The average solar irradiance in LEO is 1367 W/m2 and can reach up to 1414 W/m2 in maximum conditions. As shown in Figure 12 the first value was used as the boundary condition for the thermal analysis of EGOS-1 on SolidWorks Flow Simulation using conduction and radiation features and excluding fluid flow [4].

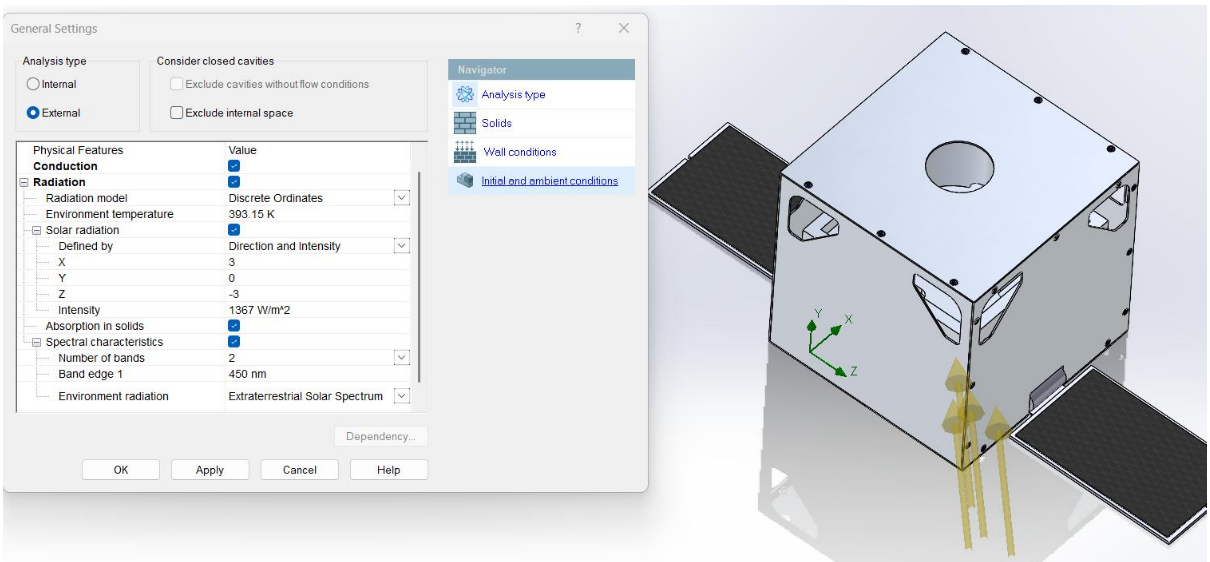


Figure 12 Boundary conditions of the thermal simulation of EGOS-1 showing the arrows as the sun exposure on the frame with the initial solar irradiance value.

6.1.5.3 Modal Analysis

To evaluate the dynamic behaviour of the frame, a modal analysis was conducted to determine its natural frequencies and mode shapes. This ensures that the structure's natural frequencies do not coincide with the rocket excitation frequencies, thereby preventing resonance and ensuring structural integrity during launch [9], [10], [13].

The simulation was conducted on ANSYS Student, that caused limitation in the number of mesh elements therefore the mesh quality was really low at an element size of 5 cm. The boundary condition was a fixed support at the bottom plate to simulate the attachment to the rocket shown in Figure 13.

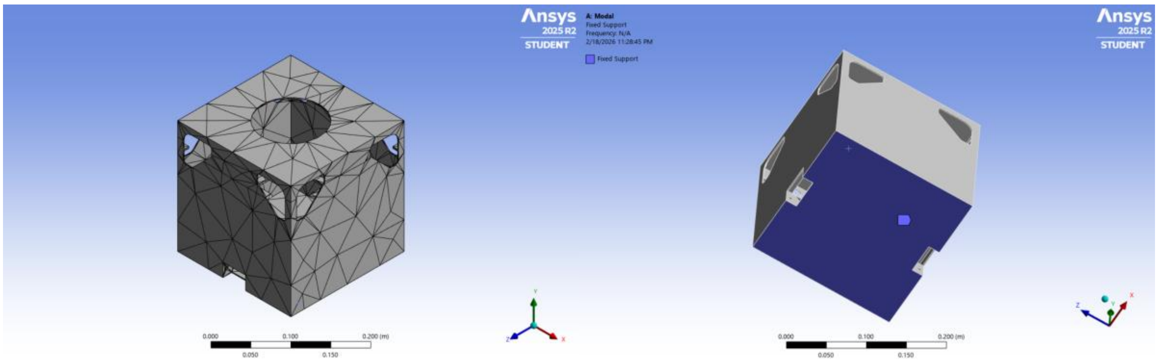


Figure 13 Mesh quality plot for the modal analysis (the left image), and the fixed support of the bottom plate as a boundary condition (the right image).

6.1.6 Simulations Results

6.1.6.1 Static Simulation Results

The static structural analysis was performed to evaluate the ability of the microsatellite frame to withstand launch loads and ensure structural integrity during the most critical phase of the mission. A 10G acceleration load was applied to simulate the mechanical stresses experienced during launch. The results obtained from the simulation indicate that the structure experiences a maximum deformation of 0.026 mm, which is extremely small relative to the overall dimensions of the satellite frame (20 × 20 × 20 cm). This minimal deformation suggests that the structural configuration of the aluminum frame is sufficiently rigid to resist mechanical loading without experiencing significant displacement or structural distortion.

The maximum equivalent stress observed in the structure was 39.371 MPa. When compared with the yield strength of Aluminum 6061-T6 (275 MPa), the stress value remains well below the material's allowable limit. This indicates that the material operates far from its failure threshold under the simulated loading conditions. As a result, the calculated factor of safety (FOS) of approximately 7 demonstrates a high structural margin against yielding or permanent deformation. Such a safety margin is desirable in space structures where reliability and robustness are critical due to the inability to perform repairs once deployed in orbit.

The deformation, stress, and factor-of-safety plots presented in Figure 14 further confirm that the stresses are distributed within acceptable limits throughout the frame. No localized stress concentrations exceeding the material capacity were observed. Therefore, the results indicate that the designed microsatellite frame can safely tolerate the expected launch loads while maintaining structural integrity.

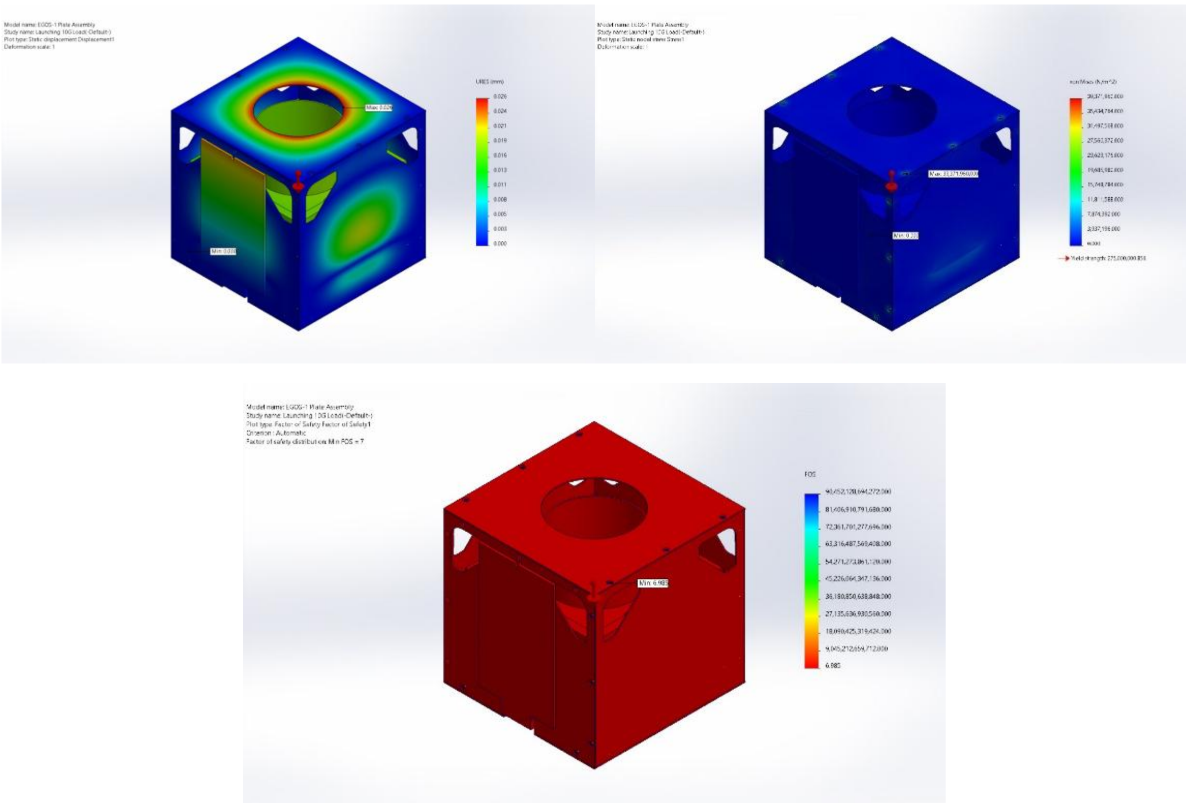


Figure 14 Deformation plot, Stress Plot, FOS Plot

6.1.6.2 Thermal Simulation Results

Thermal analysis was conducted to evaluate the temperature distribution across the satellite frame under orbital environmental conditions, primarily considering exposure to solar radiation. The simulation results indicate that the maximum temperature reached by the structure is 411.51 K (approximately 139°C). Although this represents a relatively high temperature compared to typical terrestrial operating conditions, it remains within the acceptable operational limits of Aluminium 6061-T6, whose melting point is approximately 855 K (582°C).

The temperature distribution shown in Figure 15 indicates that the thermal loading is not uniform across the satellite structure. Regions directly exposed to solar radiation experience higher temperatures, while shaded areas remain relatively cooler. However, even at the peak temperature value, the structure remains well below the material's thermal failure limits, indicating that the frame can tolerate the expected thermal environment in low Earth orbit.

It is important to note that the analysis focuses primarily on the structural material rather than sensitive electronic components or payload elements. While the structural frame can withstand these temperatures, additional thermal control strategies such as surface coatings, insulation, or passive radiators may be required in a detailed mission design to ensure that onboard electronics and sensors operate within their recommended temperature ranges.

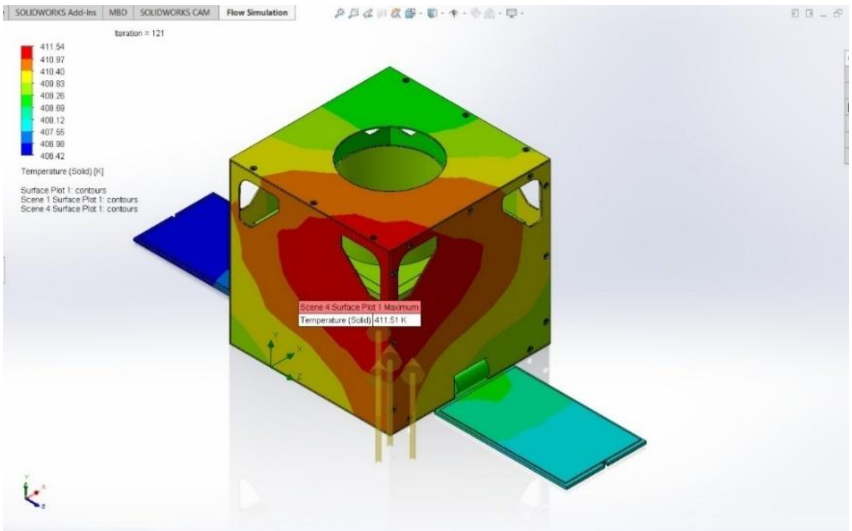


Figure 15 Temperature plot on the frame showing a maximum value of 411.51 K.

6.1.6.3 Modal Analysis Results

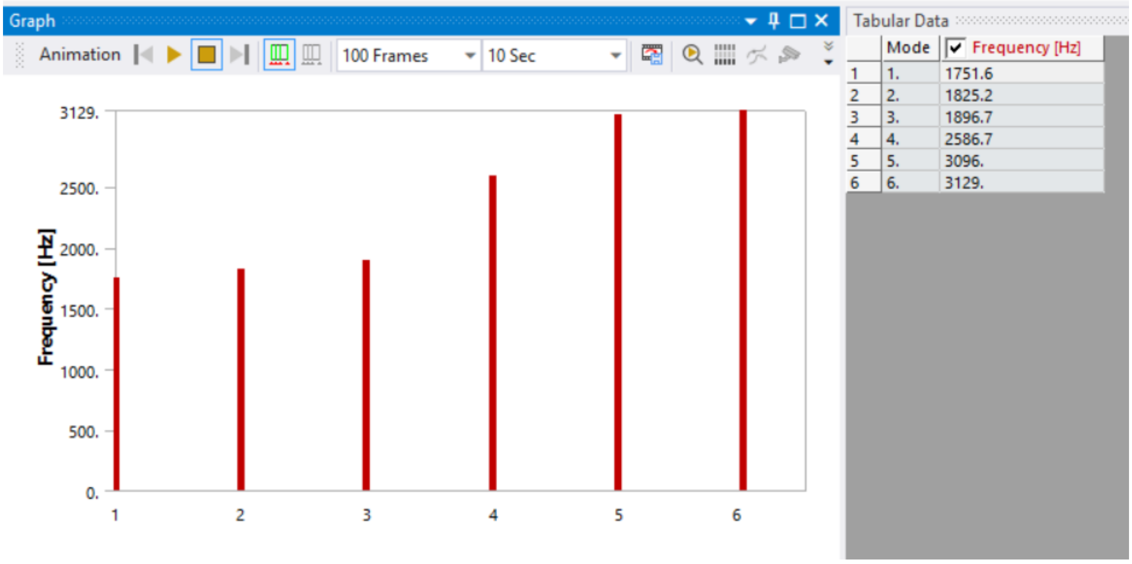
Modal analysis was performed to evaluate the dynamic behaviour of the microsatellite structure and determine its natural frequencies and vibration modes. This analysis is particularly important for launch conditions, where intense vibrations from the launch vehicle can potentially excite the natural frequencies of the satellite structure and lead to resonance.

A total of six vibration modes were analysed, and the corresponding natural frequencies were determined, as presented in Figure 16. The associated deformation shapes for each vibration mode are illustrated in Figure 17. These mode shapes describe the different ways in which the satellite structure may oscillate when subjected to dynamic excitation.

The results show that the structure possesses distinct natural frequencies for each mode, indicating predictable and stable dynamic behaviour. The deformation patterns in the modal shapes demonstrate that the structure maintains its integrity without excessive displacement under vibrational excitation. Importantly, the identified natural frequencies are expected to remain sufficiently separated from the dominant frequency ranges typically produced during launch vehicle vibrations. This separation reduces the probability of resonance occurring during launch.

The modal deformation plots further indicate that the largest displacements occur primarily at the outer structural elements of the frame, which is consistent with expected behaviour in lightweight frame structures. However, these displacements remain within acceptable limits and do not indicate structural instability.

Overall, the modal analysis confirms that the designed microsatellite frame exhibits adequate dynamic stiffness and stability, making it suitable for surviving launch-induced vibrations without structural failure or resonance-related amplification.



	Mode	Frequency [Hz]
1	1.	1751.6
2	2.	1825.2
3	3.	1896.7
4	4.	2586.7
5	5.	3096.
6	6.	3129.

Figure 16 Frequency values in the 6 modes.

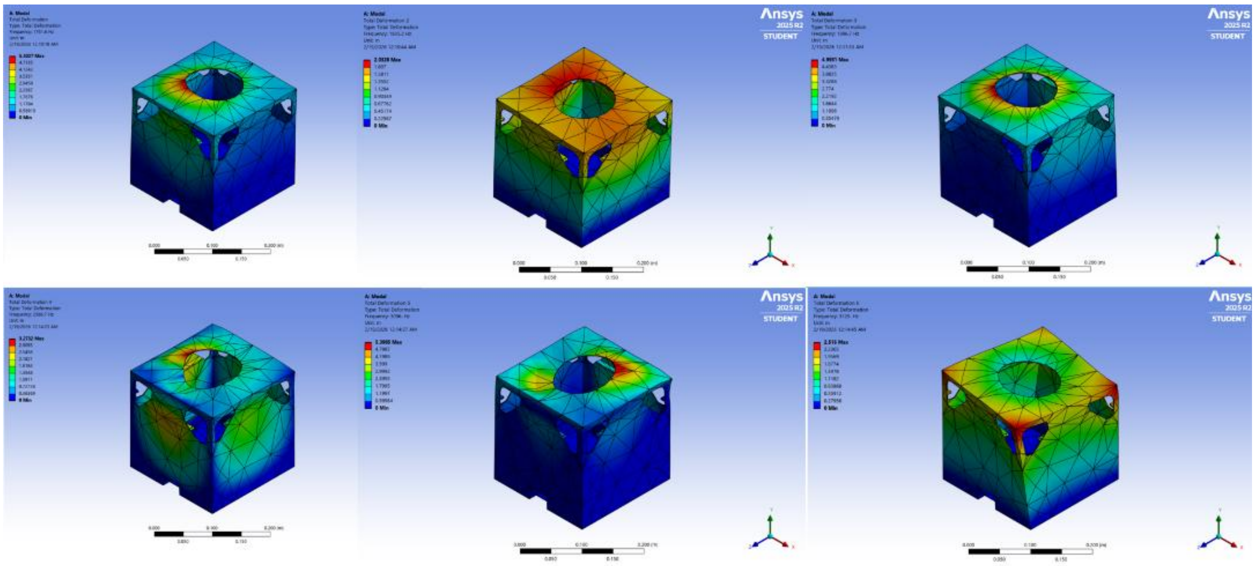


Figure 17 Total deformation plot in each of the 6 modes at different frequencies.

6.1.7 Discussion & Summary

The combined structural, thermal, and modal analyses provide a comprehensive evaluation of the proposed microsatellite frame design. The static analysis confirms strong structural robustness, with stress levels well below the material limits and minimal deformation under simulated launch loads. The thermal analysis demonstrates that the structure can tolerate the orbital thermal environment, although additional thermal management may be required to protect sensitive electronic components. Finally, the modal analysis indicates that the structure possesses adequate dynamic stiffness and stable vibration characteristics, reducing the likelihood of resonance during launch.

Taken together, these results suggest that the proposed microsatellite frame design meets the fundamental structural and environmental requirements for operation in low Earth orbit. While the analyses indicate satisfactory performance under the simulated conditions, further refinement through detailed subsystem integration, mass optimization, and full mission-level environmental testing would be required before final deployment.

EGOS-1 Static Simulation	
Maximum Displacement	0.026 mm
Maximum Stress	3.937 MPa (Y.S=275 MPa)
FOS	7

EGOS-1 Thermal Simulation	
Maximum Temperature	411.51 K

EGOS-1 Modal Analysis	
Mode 1	56.337 Hz (Safe)
Mode 2	56.814 Hz (Safe)
Mode 3	333.49 Hz (Safe)
Mode 4	334.19 Hz (Safe)
Mode 5	371.2 Hz (Safe)

Figure 18 Results summary of the static, thermal and modal analysis of EGOS-1 microsatellite.

6.2 AOCS Testbed

6.2.1 Design Parameters

The design parameters of the AOCS testbed are derived from two sources: the design of the microsatellite (which set the performance envelopes that the testbed must be able to reproduce and validate), and the physical constraints of the laboratory environment, available materials, and project budget. The parameters are organised into three groups: Helmholtz cage magnetic field simulation, spherical air bearing mechanical platform, and overall AOCS performance targets.

6.2.1.1 Helmholtz Cage Design Parameters

The primary design target for the Helmholtz cage is to generate a uniform magnetic field of B = 100,000 nT (0.1 mT / 1 Gauss) — well above the maximum Earth's magnetic field at LEO (approximately 0.3 to 0.6 Gauss) — to provide full coverage of the EGOS-1 operating range. The design is based on the following governing equation for the magnetic flux density at the centre of a square Helmholtz coil pair:

$$B = \frac{(2\mu_0 NI)}{\pi a} \times \frac{2}{(1 + \gamma^2)\sqrt{2 + \gamma^2}}$$

where $\mu_0 = 12.57 \times 10^{-7}$ T·m/A (the permeability of free space), N is the number of turns per coil, I is the current through the coil, a is the half-side-length of the square coil, and

$\gamma = 0.5445$ is the optimum coil separation-to-side-length ratio that maximises field uniformity by eliminating second-order spatial gradient terms at the cage centre. Three rectangular coil pairs are designed for the three orthogonal axes (XZ, XY, and YZ planes) of the cage, each with different dimensions to accommodate the rectangular workspace, the design parameters values are shown in Table 2.

Table 2 Helmholtz cage design parameters values for each axis pair

Axis Plane	Side Length 2a (m)	Separation 2b (m)	N (turns)	Operating Current
XZ plane	0.75 m	0.408375 m	5	5 A (test) / 10 A (max)
XY plane	0.714 m	0.388773 m	5	5 A (test) / 10 A (max)
YZ plane	0.678 m	0.369171 m	5	5 A (test) / 10 A (max)

6.2.1.2 Spherical Air Bearing Design Parameters

The spherical air bearing is designed to support a satellite mock-up with payload geometry of 0.30 m cube side length and payload mass of 15 kg (weight W = 15 × 9.81 = 147.2 N). It uses the aerostatic principle: compressed air at 4–6 bar gauge is fed through a 1.0 L plenum and expelled through 13 × 1.0 mm orifices distributed over the hemispherical bowl surface, creating a thin 20 µm air film between the sphere and bowl that eliminates contact friction. The bearing components are 3D printed from PLA, with key fastening via 6 ×150 mm screws to grip on the microsatellite.

Table 3 Spherical air bearing final design summary

Component	Function	Final Specification	Material
Hemisphere	Load-bearing ball	Ø300 mm, M8 boss	PLA
Bowl	Air-bearing seat	Inner R = 150 mm, depth ~150 mm	PLA
Air Film Gap	Frictionless interface	20 µm nominal (10–50 µm range)	—
Orifice Set	Air feed holes	13 × 1.0 mm (upgradeable to 1.25/1.5 mm)	PLA
Plenum	Pressure reservoir	1.0 L volume, sealed behind bowl	PLA
Supply Port	Air inlet	¼" BSP or 8 mm push-fit	Brass/steel
Mount Interface	Satellite attachment	6 x 150 mm threaded stud	Steel or AL6061

Air Supply System	Compressor + filter	10 bar, 15 L tank; 5 µm filter, 4500 L/min	Off-the-shelf
Pedestal / Stand	Bearing mount	30 x 30 x 60 cm	MDF Wood

6.2.2 Design Environment

The AOCS testbed was designed and evaluated using an integrated hardware–software environment combining commercial off-the-shelf (COTS) components with custom-designed mechanical parts and in-house firmware. The environment is structured around modularity, allowing each testbed subsystem (Helmholtz cage, air bearing, sensor suite, control computer) to be designed, tested, and refined independently before integration.

6.2.2.1 Software Tools

MATLAB/Simulink (R2023a) — Primary platform for AOCS algorithm development, attitude dynamics modelling, and Model-in-the-Loop (MIL) simulation. The Aerospace Toolbox provides orbital mechanics, reference frame transformations (ECI, LVLH, body frame), and the IGRF-13 geomagnetic field model for generating realistic field profiles along the EGOS-1 orbit.

Wolfram Demonstration Project — Used for interactive Helmholtz cage magnetic field visualisation and uniformity analysis. This tool allowed rapid parametric exploration of coil geometry (side length, spacing, turns, current) and direct visual confirmation of the uniform field region dimensions before fabrication.

SolidWorks (2023) with Flow Simulation add-in — Used for 3D CAD modelling of both testbed assemblies and for the Computational Fluid Dynamics (CFD) simulation of the spherical air bearing aerostatics.

ANSYS Fluent (2025 R1) — Used for an independent, higher-fidelity CFD validation of the air bearing flow field under the specified boundary conditions (inlet gauge pressure 600,000 Pa, environment pressure 101,325 Pa).

6.2.3 CAD Model & Schematics

6.2.3.1 Helmholtz Cage CAD Model

The Helmholtz cage is modelled as three orthogonal rectangular coil-frame pairs mounted on a common outer support structure. The outer support structure is fabricated as an Aluminum rectangular stand, chosen for its modular assembly and non-magnetic properties. Each coil frame is built from 18 mm Aluminium U-channel, into which 5 turns of 4 mm^2 wire are wound and retained. The three coil-pair planes (XZ, XY, YZ) are positioned and separated according to the $\gamma = 0.5445$ spacing ratio derived from the magnetic field governing equation, yielding separations of 0.408375 m, 0.388773 m, and 0.369171 m respectively.

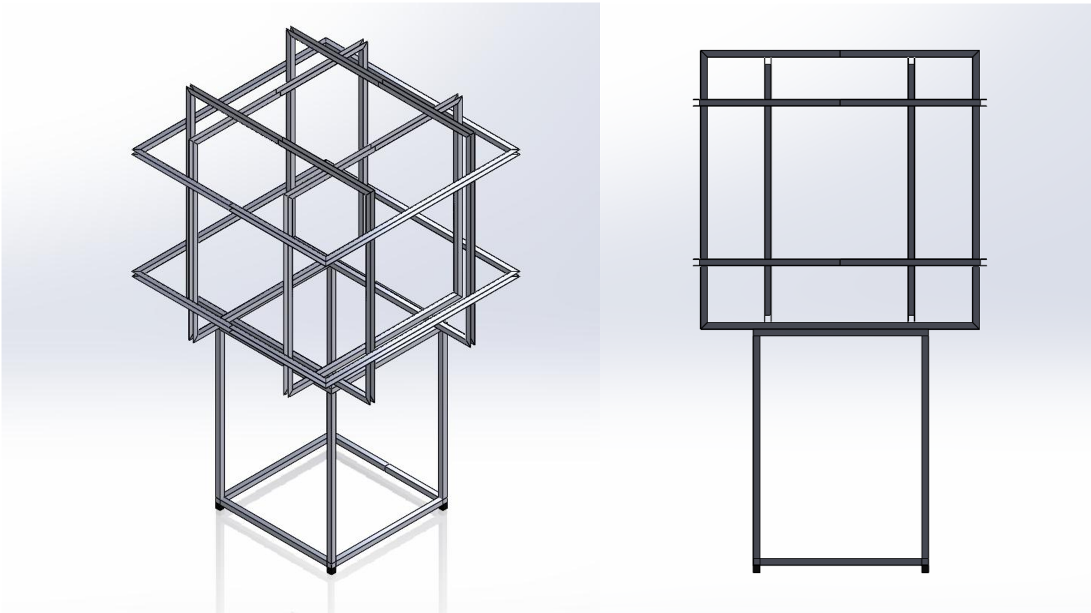


Figure 19 3D CAD model performed on SOLIDWORKS of the helmholtz cage structure & stand, showing the isometric view and front view.

6.2.3.2 Spherical Air Bearing CAD Model

The spherical air bearing CAD model captures the complete assembly: the 300 mm diameter inner sphere, the hemispherical bowl (inner radius R = 150.000 mm, depth ~150 mm, with 13 × 1.0 mm orifices) with 4 x 6 mm holes to mount the specimen on the hemisphere using a grip mechanism, the inlet fitting is a ¼" BSP.

The complete assembly is mounted on an MDF wood pedestal (30 cm diameter) at a height that centres the sphere and mounted satellite within the Helmholtz cage working volume. The total height of the air bearing assembly on the pedestal is designed to align the satellite CoM with the geometric centre of the Helmholtz cage.

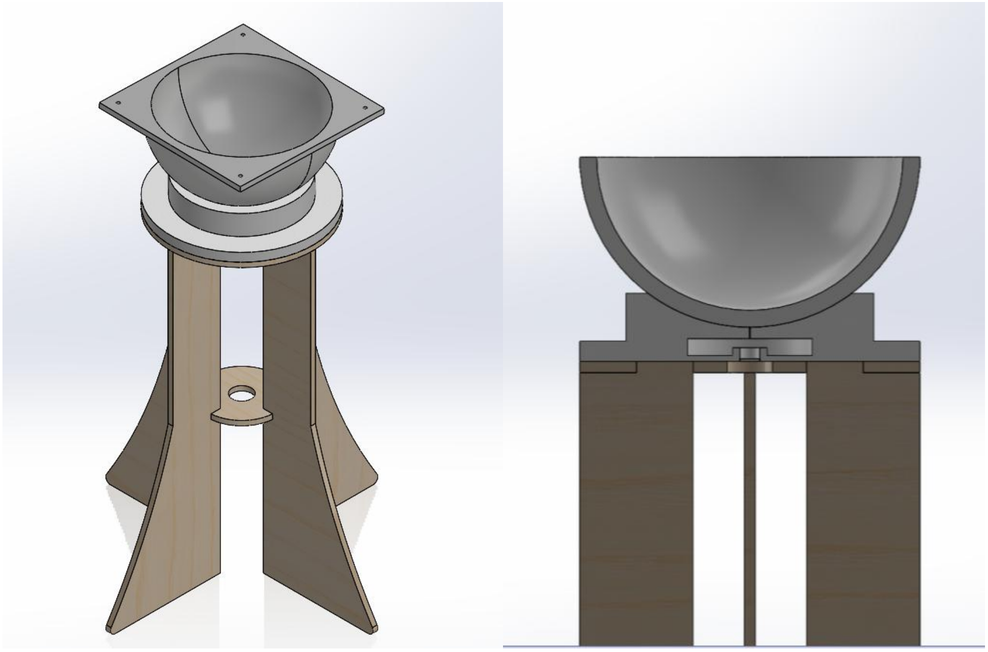


Figure 20 Spherical air bearing 3D CAD model on SOLIDWWORKS, showing the isometric view and the sectional front view to illustrate the air chamber and the central orifice.

6.2.3.3 Complete AOCS Testbed 3D CAD Model

Integerating all subsystems together, in addition to the EGOS-1 microsatellite we will come to this 3D CAD model illustrated in Figure 21.

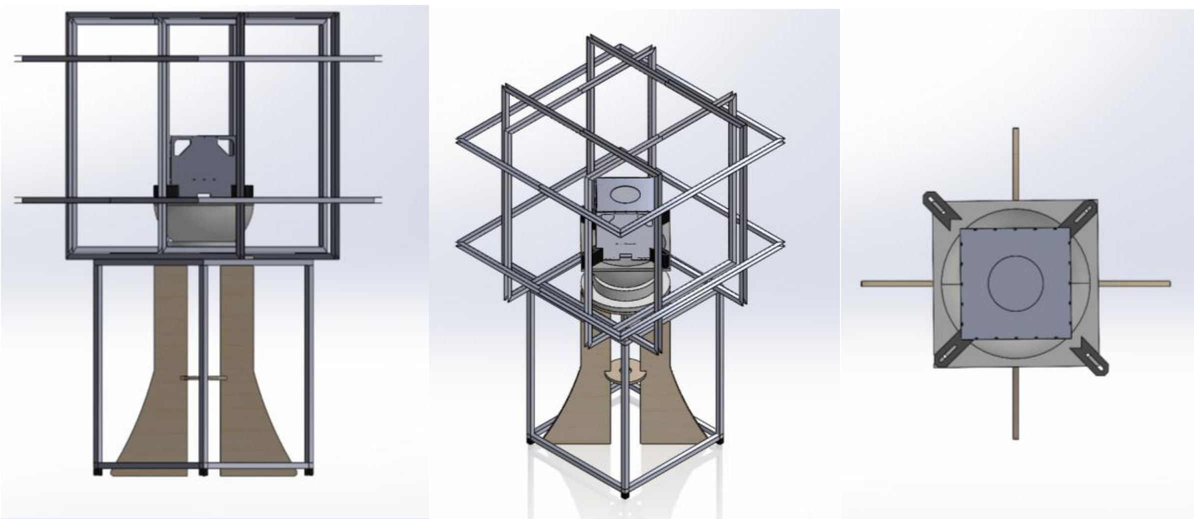


Figure 21 Full system 3D CAD model showing 2 various views, and the top view to illustrate the gripping mechanism on the microsatellite, showing 2 diagonal tight grips and 2 loose grips.

6.2.4 Simulations Setup

Two independent simulation analyses were performed to validate the testbed design: (1) magnetic field uniformity simulation of the Helmholtz cage using the Wolfram Demonstration tool, and (2) CFD simulation of the air bearing using SolidWorks Flow Simulation and ANSYS Fluent 2025 R1 in parallel.

6.2.4.1 Helmholtz Cage Magnetic Simulation Setup

The Wolfram Demonstration tool computed the axial magnetic flux density profile B_z(0,0,0) and the field distribution in the axial (z) and transverse (y) directions for each of the three cage axis planes. Input parameters: coil side lengths 0.75 m / 0.714 m / 0.678 m, separations per γ = 0.5445, N = 5 turns, I = 5 A.

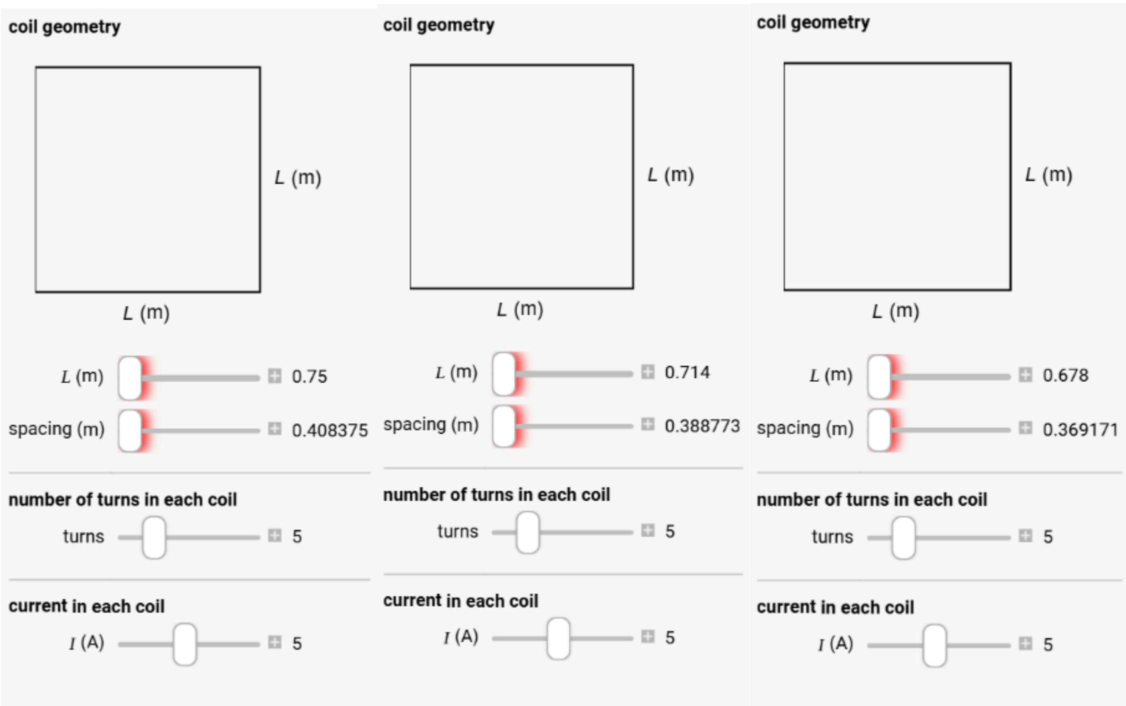


Figure 22 The input parameters for the Helmholtz cage simulation to solve for the magnetic flux density and the homogenious distance.

6.2.4.2 Air Bearing CFD Setup

In SolidWorks Flow Simulation, the bearing geometry was imported from the CAD model. Boundary conditions: total pressure inlet = 600,000 Pa (6 bar gauge), environment pressure outlet = 101,325 Pa. Simulation goals: lifting force on sphere (GG Force Y), average total pressure, shear stresses, mass flow rate, and outlet velocity. Convergence at 85 iterations.

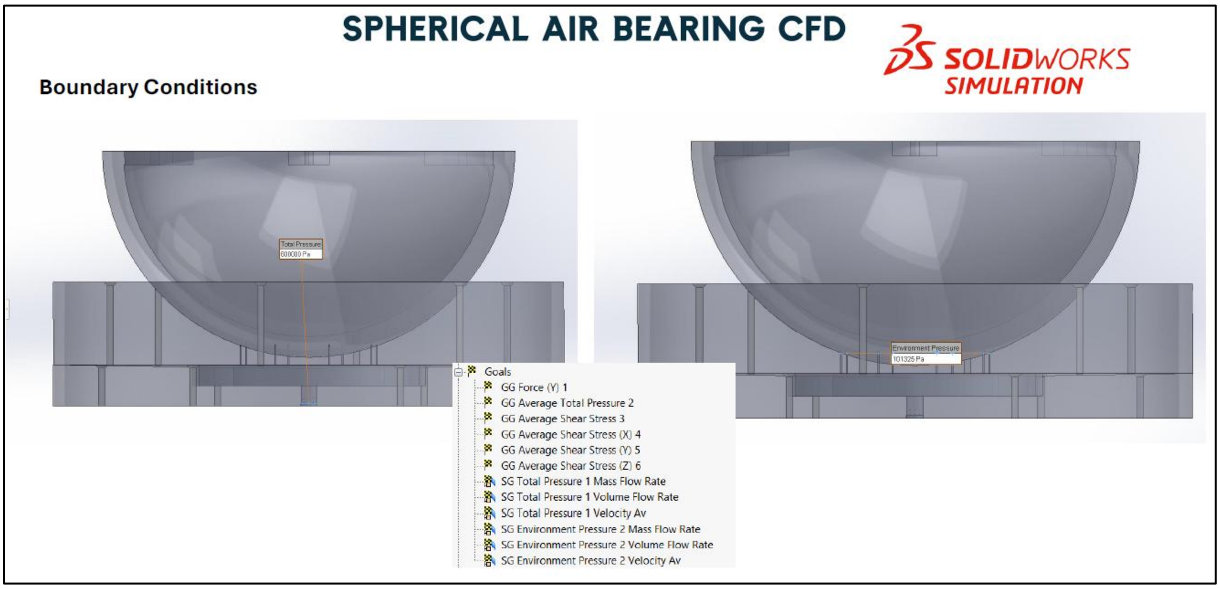


Figure 23 SOLIDWORKS Flow Simulation boundary conditions for the spherical air bearing: inlet total pressure 600,000 Pa (left) and environment pressure outlet 101,325 Pa (right). Simulation goals are listed in the centre panel.

In ANSYS Fluent 2025 R1, the same geometry was independently modelled with a pressure-inlet boundary (gauge total pressure = 600,000 Pa) and atmospheric outlet. Solver: pressure-based, steady-state, k–ε turbulence model.

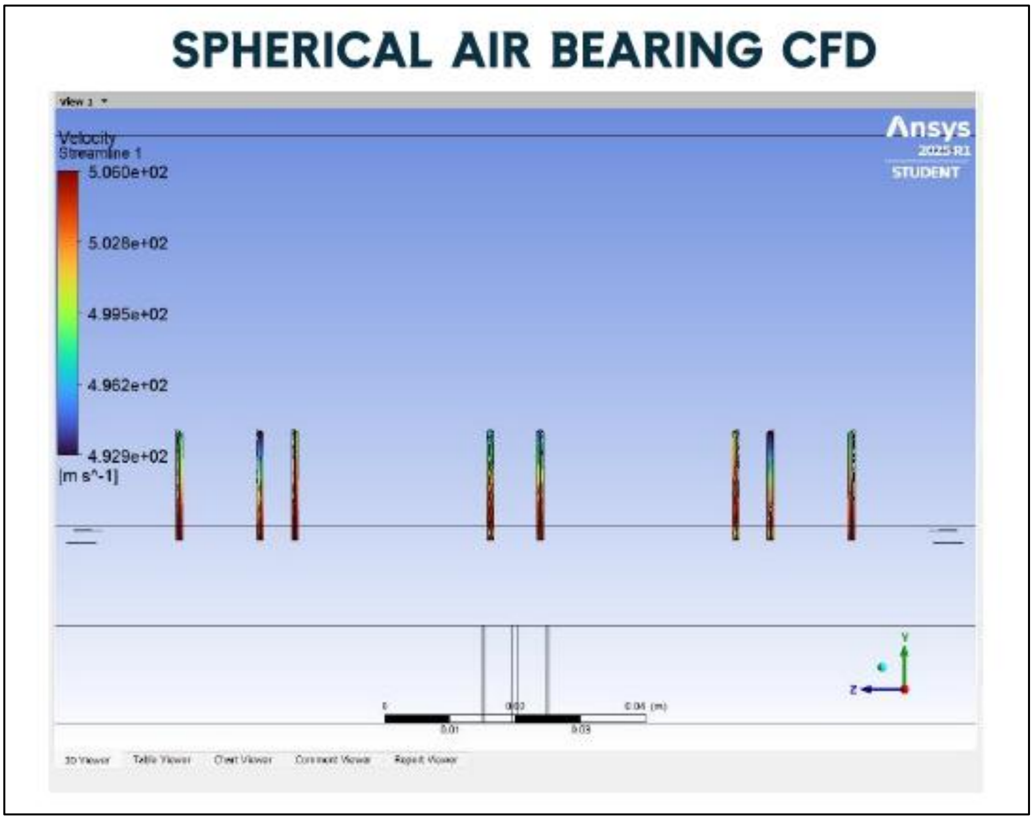


Figure 24 ANSYS Fluent 2025 R1 setup for the spherical air bearing CFD simulation: bowl geometry with pressure inlet at 600,000 Pa gauge (normal to boundary), and velocity streamline initialisation showing the eight orifice inlet jets.

6.2.5 Simulations Results

6.2.5.1 Helmholtz Cage Magnetic Field Results

The Wolfram simulation produced axial field profiles for all three cage planes at I = 5 A. All three planes generate fields within the Earth's LEO magnetic field range (0.3–0.6 Gauss), confirming the design is correct for AOCS sensor testing at the specified operating current. The Earth's magnetic field range of 0.3–0.6 Gauss is achieved at 5 A, while the maximum designed field of 1 Gauss is achieved at 10 A.

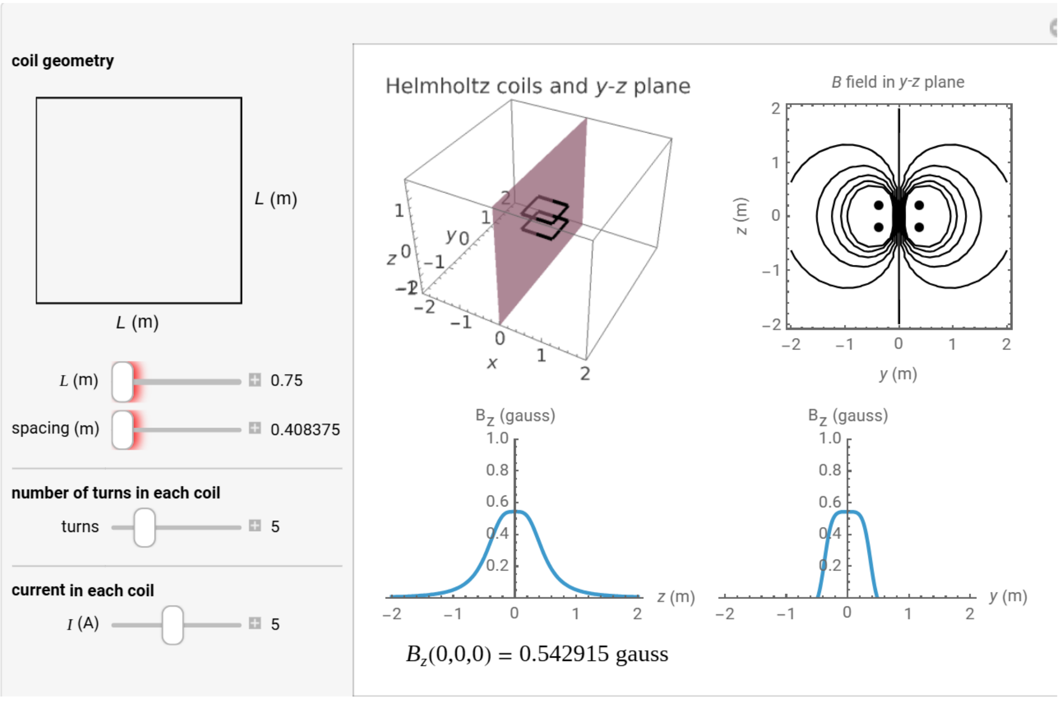


Figure 25 Wolfram Demonstration magnetic simulation for the XZ plane (2a = 0.75 m, N = 5, I = 5 A): B_z vs z (left) and B_z vs y (right) profiles, confirming B_z(0,0,0) = 0.542915 Gauss and a broad uniform-field plateau.

Table 4 Helmholtz cage magnetic simulation results: central field and uniform region for each axis plane at I = 5 A.

Plane	B_z(0,0,0) [Gauss]	B_z(0,0,0) [nT]	Uniform Field Region
XZ	0.542915	54,2915	± ~0.45 m from centre
XY	0.571606	57,1606	± ~0.45 m from centre
YZ	0.600569	60,0569	± ~0.45 m from centre

6.2.5.2 Spherical Air Bearing CFD Results

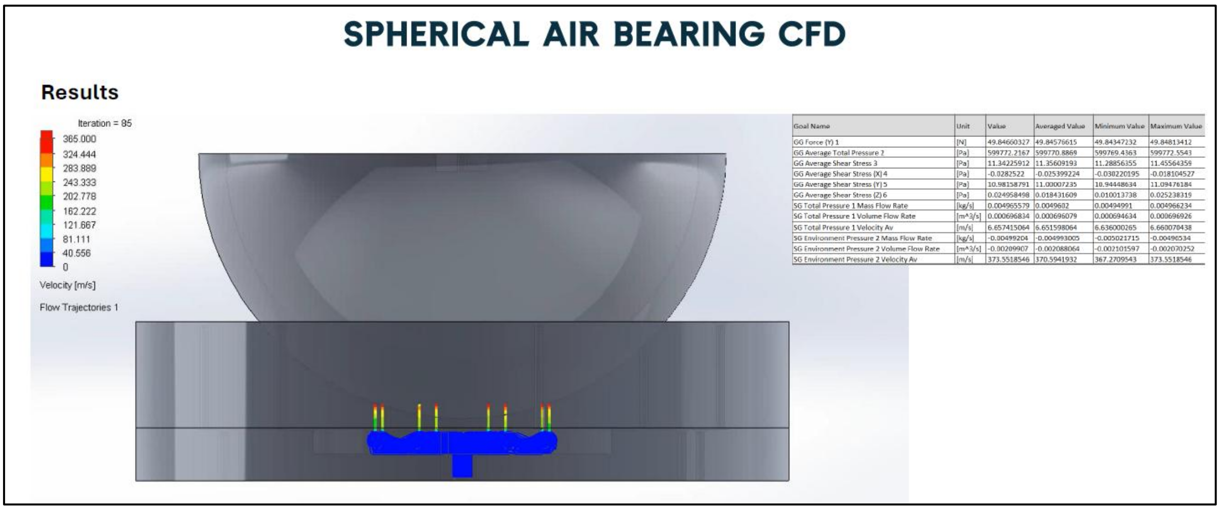


Figure 26 SolidWorks Flow Simulation results at iteration 85: velocity flow trajectories (left, scale 0–365 m/s) showing symmetric air film formation between the orifice jets and the sphere surface. The results table (right) shows GG Force Y = 49.845 N (averaged)

Table 5 SolidWorks Flow Simulation results for the spherical air bearing (6 bar inlet, 13 × 1.0 mm orifices, 85 iterations).

Goal / Output	Value (Averaged)	Unit
GG Force Y (lifting force on sphere)	49.845	N
GG Average Total Pressure (bowl surface)	599,770.89	Pa
GG Average Shear Stress (total)	11.356	Pa
SG Total Pressure — Average Velocity	6.658	m/s
SG Total Pressure — Mass Flow Rate	0.00490	kg/s
SG Environment Pressure — Avg Outlet Velocity	370.594	m/s

Table 6 ANSYS Fluent 2025 R1 CFD results for the spherical air bearing. Both tools agree on the lifting force to within < 0.01%.

Result	Value	Unit
Net upward thrust (lifting force)	49.85	N
Outlet maximum velocity (orifice jet)	506	m/s

6.2.6 Discussion

6.2.6.1 Helmholtz Cage

The simulations confirm that the three-axis rectangular Helmholtz cage with N = 5 turns and γ = 0.5445 spacing generates the Earth's magnetic field range (0.3–0.6 Gauss) at I = 5 A — well within standard laboratory DC power supply ratings. The maximum field of 1 Gauss at 10 A provides a 3× margin above the Earth's field upper bound, enabling sensor saturation and over-range calibration tests. The uniform field region of 0.45 m radius from the cage centre comfortably encloses the EGOS-1 mock-up (maximum dimension 20 cm on the air bearing), providing 12.5 cm clearance margin on each side. The slightly different per-axis field strengths (0.542915, 0.571606, and 0.600569 Gauss at 5 A) result from the rectangular geometry and are compensated by adjusting each axis current independently, which is standard Helmholtz cage practice.

6.2.6.2 Spherical Air Bearing

Both SolidWorks and ANSYS Fluent agree on a lifting force of 49.85 N against the required 147.2 N. This shortfall is fully explained by choked-flow physics: at a supply-to-ambient pressure ratio of 600,000 / 101,325 ≈ 5.9 (above the critical ratio of ~1.89 for air), the orifice mass flow is at its maximum for 13 × 1.0 mm holes. The outlet jet velocities of 370–506 m/s confirm choked flow. The strong agreement between the two independent CFD tools validates the simulation methodology.

7. Deliverables & Implementation

7.1 Space Observatory Microsatellite

For the implementation of the EGOS-1 microsatellite, the hardware components, including the microsatellite frame, electronic systems, and circuit assemblies, were designed, modelled, and fabricated according to the previously specified dimensions and schematics. However, due to fabrication limitations associated with the payload subsystem, the scintillating crystal could not be manufactured. Consequently, the payload implementation was excluded from the final prototype development.

Additionally, an issue was encountered with the dimensions of the supplied solar panels. Due to stock unavailability, the supplier provided panels with dimensions slightly larger than those originally specified in the order. As a result, the solar panels were incompatible with the designed mounting base and could not be properly aligned or integrated into the microsatellite structure. Consequently, the solar panels were excluded from the final implementation.

7.1.1 Hardware Components

The hardware components are categorized into two main subsystems: the electronic components and the structural frame.

7.1.1.1 Electronic Components

The electronics in any microsatellite are divied to subsystems depending on there functionality, the subsystems are:

- Electrical Power Subsystem (EPS)
- On-Board Computer (OBC)
- Attitude & Orbit Control System (AOCS)
- Communication System (COMMS)

The following *Table 7* illustrates the final installed components and there modules.

Table 7 Final installed electrical components for the EGOS-1 trial model.

Electrical Power Subsystem (EPS)		
Component	***Module***	***Function***
3S2P 12 V Li-Ion Battery	18650 12V 3S-2P+BMS 6000mAh	Power Source
DC Power Distributor 4-Channel Module	Power Distributor Module for 5–48V Systems With Per-Channel 10A Outputs	Multiple Power Outputs
2x Adjustable DC-DC Step-Down Converter	XL4015 Adjustable DC-DC Step-Down Module with Display 5A 75W	Controls voltage output to match the input for other devices.
Onboard Computer (OBC) + Communciation System (COMMS)		
Component	***Module***	***Function***
ESP32 Dual-Core MCU (240 MHz)	ESP32 DevBoard	Data management and controlling other systems, and telemetry acquisition.
Attitude & Orbit Control System (AOCS)		
Component	***Module***	***Function***
Brushless DC Motor — Reaction Wheel	A2212 BLDC Motor	Constant correction of the satellites attitude.
Electronic Speed Controller (ESC)	40A ESC	Drives the motor with precise current.

Custom Magnetorquers ×2	240-turn coils of 0.4 mm enamelled copper wire	Interacts with Earth's field B to produce torque $\tau = m \times B$ about the pitch and roll axes.
Dual H-Bridge Driver	L298N	Controls current magnitude and direction in each magnetorquer coil for pitch and roll control.
IMU	MPU-6050	Six-axis MEMS sensor (gyroscope + accelerometer) providing three-axis angular rate feedback via I^2C for the real-time control loop.
Magnetometer	HMC5883L	Three-axis magnetic field sensor (up to 160 Hz, I^2C) for attitude determination and magnetorquer feedback control.

Listing all of the components and there function, and studying there data sheet will lead to the step of making the final schematic for the mictrosatellite model. The purchased components and the final implemented schematic is shown in Figure 27.

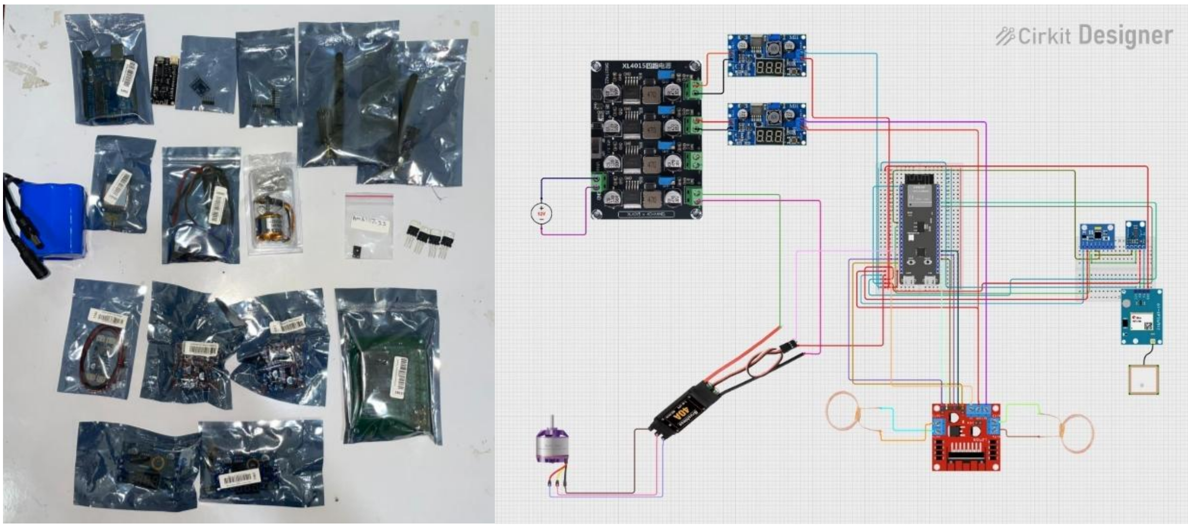


Figure 27 The final purchased components (right image) & The final schematic (left image).

7.1.1.2 Structural Frame Fabrication

The microsatellite frame consists of a hollow cubic structure with overall dimensions of 20 × 20 × 20 cm and includes a circular hatch with a diameter of 10 cm located on the top surface, as previously illustrated in

Figure *8*. The structure is composed of six individual plates: two front plates, two side plates, one top plate, and one bottom plate. Each plate has a thickness of 1.5 mm, and when assembled, they form the final cubic configuration.

The selected manufacturing method for the frame components was CNC laser cutting. Initially, Aluminum T6061 was considered as the primary construction material due to its lightweight properties and high thermal conductivity, which make it suitable for heat dissipation applications in space systems. However, due to material availability and manufacturing convenience, Stainless Steel 304 (SS304) was utilized instead. Although SS304 has a significantly higher density, approximately three times heavier than Aluminum T6061, it was considered acceptable for the fabrication of the Earth-based trial model.

Prior to the CNC laser cutting process, final modifications were implemented on the design model using the Sheet Metal features in SOLIDWORKS. A tab-and-slot mechanism was incorporated into the design to facilitate the assembly and secure installation of the mounting strips onto the plates. Additionally, internal support shelves were integrated into the structure to accommodate and support the nested stack configuration as shown in Figure 28.

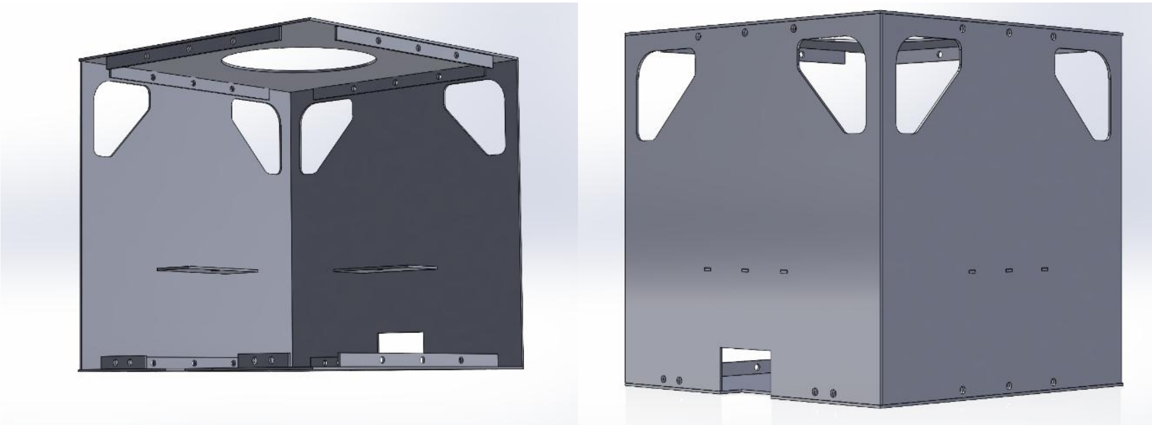


Figure 28 Plate assembly of the frame using the tab-and-slot feature to install the mounting strips that will hold the plates together, also showing the mounting shelf.

The CNC laser cutting manufacturing process was successfully completed, and the frame was fabricated and assembled with high dimensional accuracy in accordance with the finalized design specifications as shown in Figure 29.

Figure 29 The CNC laser cutting process of the structural frame plates (left image), the plates and mounting strips before assembly (middle image), the plates during assembly using 4mm button head bolts with nuts (right image.)

7.1.2 Control & Software

The software system operates on the ESP32 microcontroller, which is responsible for managing the Attitude and Orbit Control System (AOCS) and communication processes. The firmware was developed using the Arduino IDE in the C++ programming language.

The primary objective of the software is to execute the control algorithm (shown in Figure 31) responsible for acquiring state data from the inertial measurement and magnetic field sensors, namely the MPU6050 and HMC5883L modules. The collected sensor data is processed to determine the satellite's attitude and calculate the required corrective actions. These corrections are then applied through the actuators, including the reaction wheel and magnetorquers, to stabilize and control the satellite orientation.

Furthermore, the system telemetry and operational data are visualized through a real-time dashboard implemented using embedded HTML code integrated within the firmware.

7.1.2.1 Attitude Determination

The ESP32 continuously acquires angular velocity and acceleration data from the MPU-6050 sensor, in addition to three-axis magnetic field measurements from the HMC5883L magnetometer, through the I^2C communication protocol at fixed sampling intervals. A complementary filter algorithm is implemented to fuse the gyroscope integration data with magnetometer-based tilt correction, enabling real-time estimation of the satellite's roll, pitch, and yaw orientation angles with improved stability and reduced drift.

7.1.2.2 Attitude Control — PD Controller

A Proportional-Derivative (PD) controller was implemented to compute the attitude error, defined as the difference between the measured orientation and the commanded reference angle, and to generate the corresponding actuator control signals. For yaw-axis stabilization, the controller outputs a PWM speed command to the reaction wheel electronic speed controller (ESC), while current commands are supplied to the L298N magnetorquer driver for pitch and roll attitude correction.

The controller gains were optimized using the Particle Swarm Optimization (PSO) artificial intelligence technique on the AOCS experimental testbed to achieve improved system stability, response time, and control performance.

7.1.2.3 Telemetry Dashboard

The ESP32 hosts a lightweight HTTP server through its integrated Wi-Fi access point, enabling wireless communication and monitoring capabilities. A browser-based telemetry dashboard was developed to display real-time system parameters, including attitude angles, angular velocities, magnetometer measurements, and actuator operating states.

Through the dashboard interface shown in Figure 30, operators are able to transmit attitude control commands directly to the system for testing and operational purposes. Additionally, all acquired telemetry and control data are logged into the onboard flash memory to facilitate data storage, analysis, and post-processing evaluation.

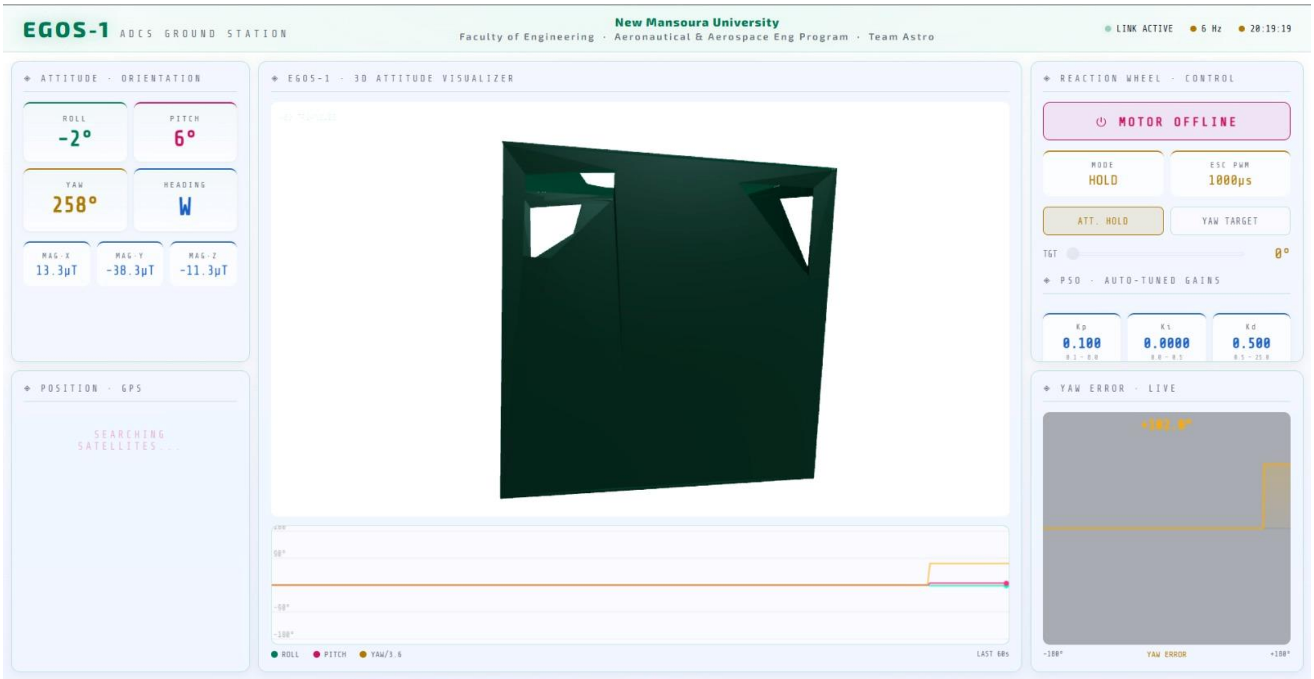


Figure 30 Telemetry dashboard for EGOS-1 microsatellite.

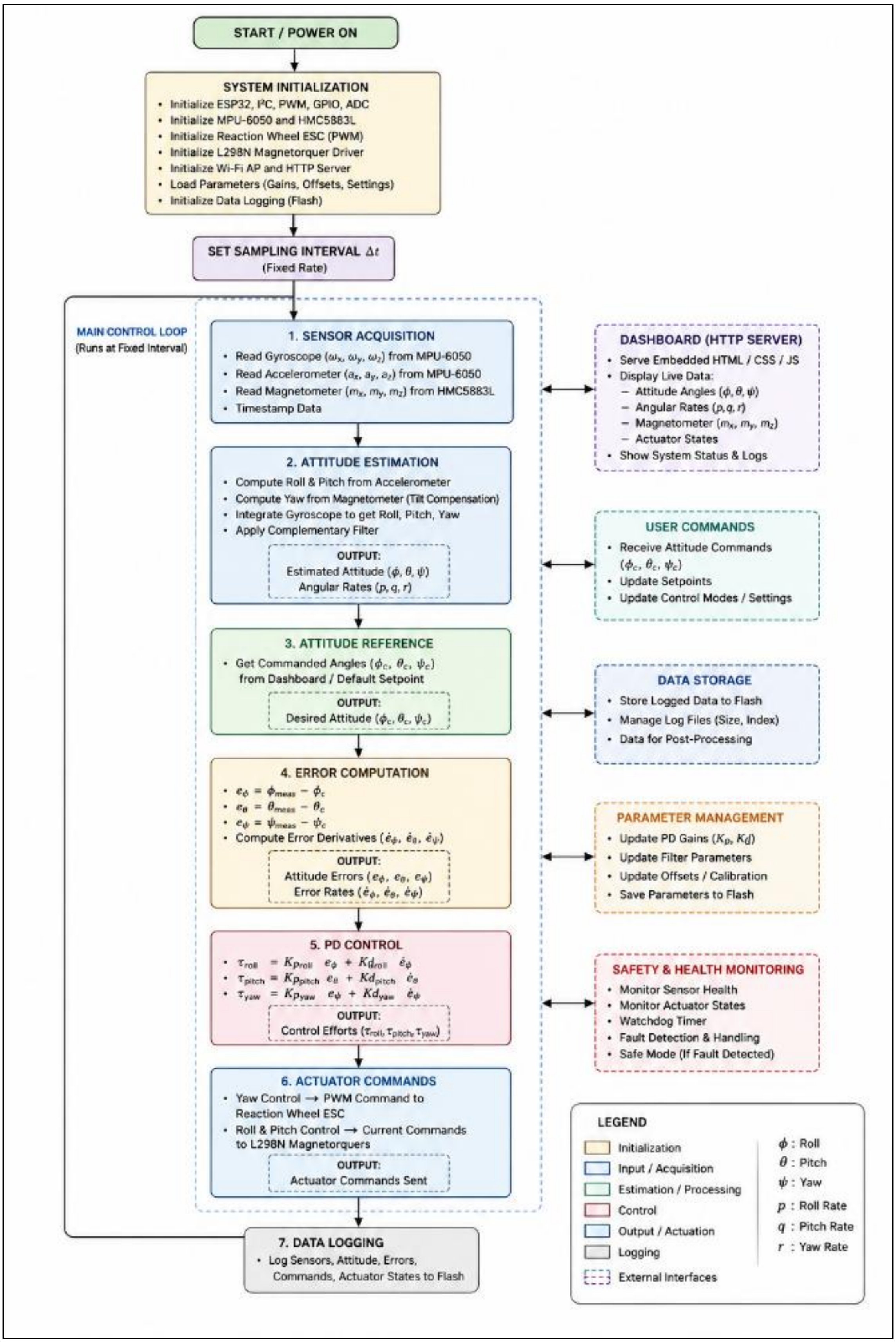


Figure 31 The control algorithim for the microsatellite's software.

7.2 AOCS Testbed

7.2.1 Hardware Components

The AOCS testbed hardware is composed of two primary mechanical subsystems: the Helmholtz cage and the spherical air-bearing assembly. These subsystems are supported by auxiliary components, including a variable power supply, pneumatic support equipment. All hardware components were selected and integrated to satisfy the magnetic field generation and rotational freedom requirements outlined in Section 6.2.

7.2.1.1 Helmholtz Cage Fabrication

Shown in Table 8 are the hardware components used to make and assemble the Helmholtz cage, showing the components in *Figure **32***.

Table 8 Hardware components for the Helmholtz cage.

Component	Specifications	Function
U-Channel Aluminum Profile	8 x 18mm x 75 cm 8 x 18mm x 71.4 cm 8 x 18mm x 67.8 cm	Support the coils winding and build the structure of the cage.
Winding Wire	4.0 mm^2 Stranded Wire – Elsewedy Electric H07V-R PVC Insulated	The coils copper wire generating the magnetic field.
Zip Ties	6 mm x 30 cm	Fasteners to hold the cage together
Stand	50 x 50 x 60 cm	Supporting the cage on the floor and centering it with the spherical air bearing.

Figure 32 The U-channel aluminum profile (left image), The 4mm2 stranded insulated wire (middle image), The zip ties used as fasteners (right image).

7.2.1.2 Spherical Air Bearing Fabrication

The fabrication of the spherical air-bearing system involved modelling the final assembly components using SOLIDWORKS, followed by manufacturing the designed parts through 3D printing using PLA material. The fabricated components were assembled and tested to ensure proper functionality and rotational performance of the spherical air-bearing mechanism. Additionally, a mounting table stand fabricated from MDF wood was constructed to provide structural support and stable installation for the spherical air-bearing assembly during testing and experimental operation. The fabrication and assembly phase are illustrated in Figure 33.

Figure 33 Final modelling of the assembly on SOLIDWORKS (left image), The 3D printing process of the parts (middle images), Final implemented assembly with stand (right image).

7.2.2 Control

The testbed control framework is operated through an adjustable power supply connected to the Helmholtz cage to regulate the generated magnetic field intensity, while an air compressor system with a controllable valve is utilized to adjust the air pressure supplied to the spherical air-bearing assembly. This configuration enables precise control of both the magnetic simulation environment and the rotational freedom conditions during experimental testing.

7.2.2.1 Helmholtz Cage — Current Control

The required current values for each Helmholtz cage axis are determined based on the simulation results presented in Section 6.2.5, which provide the expected Earth magnetic field vector components (B_x, B_y, B_z). The operator manually adjusts the corresponding power supply currents for each coil axis to reproduce the desired magnetic field conditions. This approach enables the satellite's magnetometers and magnetorquers to be evaluated and tested under a wide range of simulated orbital magnetic environments representative of real Earth-orbit conditions.

7.2.2.2 Spherical Air Bearing — Pressure Control

Pressurized air supplied by the air compressor enters the spherical air-bearing inlet and flows through the designed orifices, generating a thin air film that enables the hemispherical structure to float with minimal friction. The supplied pressure can be adjusted using the compressor regulator to achieve stable operational conditions.

Based on the simulations presented in Section 6.2.5, the recommended operating pressure range was determined to be between 2 and 4 bar. Exceeding this pressure limit may generate an unstable airflow condition and localized shockwave effects, which can disrupt the air film stability and lead to malfunction of the spherical air-bearing system.

In the following figures the operation diagram of the AOCS testbed is illustrated in Figure 34 addition to the final fabrication of the full project.

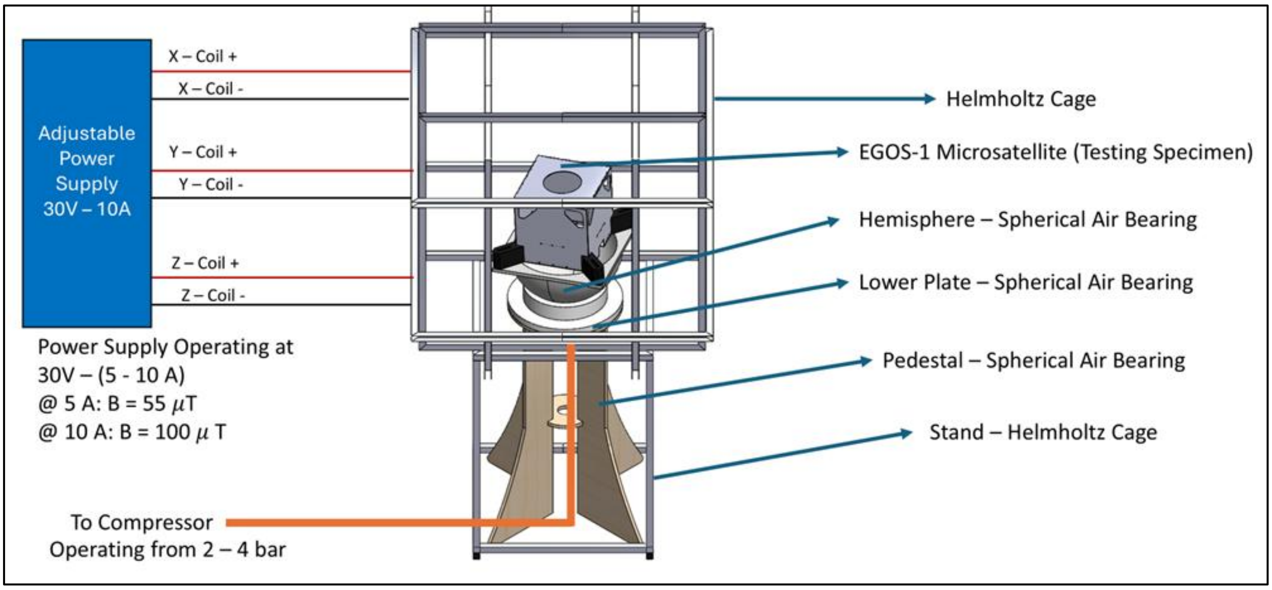


Figure 34 Testbed operation diagram

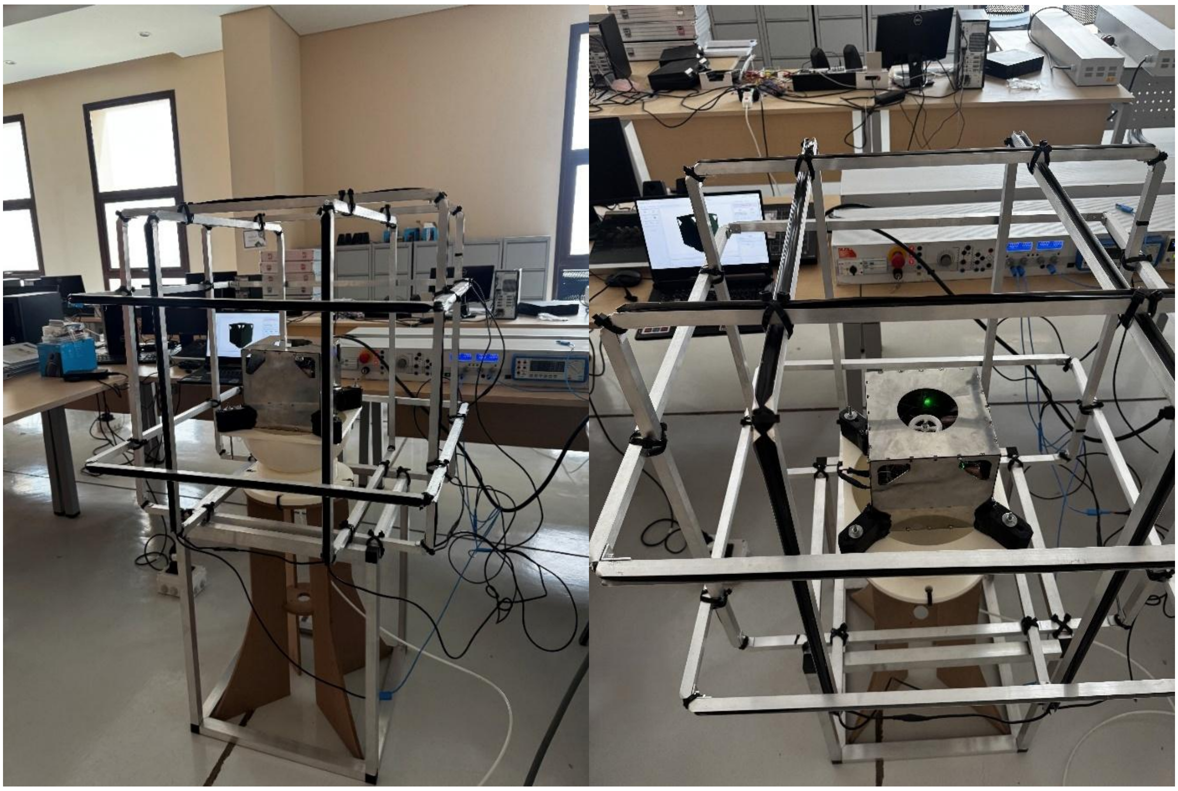


Figure 35 Full testbed + specimen project operating in a lab at New Mansoura University.

8. Possible Stakeholders

The EGOS-1 micro-satellite and AOCS Testbed project addresses a well-defined gap in Egypt's space engineering infrastructure. By combining a scientifically capable satellite platform with a domestically fabricated, low-cost test facility, the project creates tangible value for a broad and diverse set of stakeholders across government, academia, and industry as illustrated in Table 9.

Potential stakeholders were identified based on alignment with two primary project outputs: (1) the EGOS-1 micro-satellite as a scientific and educational platform, and (2) the AOCS Testbed as a replicable laboratory infrastructure product.

Table 9 Stakeholders list of the project.

Category	Stakeholder	Type	Role & Interest
Schools	Ministry of Education & Technical Education	Government	Interested in adding space and satellite engineering topics to STEM education programs.
Schools	Egyptian Japanese Schools (EJS)	Educational	Potential users of educational satellite kits and practical engineering activities.
Schools	STEM Egypt Schools	Educational	Interested in hands-on satellite experiments and STEM-based space projects.
Universities	Ministry of Higher Education & Scientific Research	Government	Supports university research, STEM development, and local space technology projects.
Universities	Academy of Scientific Research and Technology (ASRT)	Research Funding	Possible funding partner for EGOS-1 development and testing activities.

Universities	The British University in Egypt (BUE)	Private University	Potential research partner for AOCS systems and satellite engineering projects.
Universities	Zewail City of Science and Technology	Research University	Interested in astrophysics, satellite systems, and advanced space research.
Research Centers & Organizations	Egyptian Space Agency (EgSA)	Space Agency	Main national space authority supporting local satellite design, manufacturing, and testing.
Research Centers & Organizations	National Research Centre (NRC)	Research Institution	Interested in materials, engineering research, and satellite structural testing.
Research Centers & Organizations	National Authority for Remote Sensing and Space Sciences (NARSS)	Space Science	Interested in satellite data and low-cost local satellite platforms.
Research Centers & Organizations	National Research Institute of Astronomy and Geophysics (NRIAG)	Scientific Research	Interested in gamma-ray and astrophysics data from the EGOS-1 mission.
Research Centers & Organizations	Space Science Center — Cairo University (SSC/CU)	Research Center	Potential collaborator in satellite research and AOCS testbed applications.

8.1 Schools

At the school level, the project's primary value lies in the educational satellite kit concept: a simplified, low-cost derivative of the EGOS-1 Earth prototype designed for hands-on use in secondary school laboratories. The Ministry of Education and Technical Education, Egyptian Japanese Schools, and STEM Egypt Schools represent the institutional gateway to this market, through which the Astro Space Technologies startup could deploy educational products that expose students to real satellite engineering hardware and experiments at an early age.

8.2 Universities

Universities represent the core academic stakeholder group for both the AOCS Testbed and the EGOS-1 research platform. New Mansoura University is the primary beneficiary, gaining permanent laboratory infrastructure through the testbed. Broader dissemination — enabled by the project's thorough design documentation — allows other Egyptian engineering faculties to replicate the facility at low cost, gradually building a national network of AOCS research laboratories. The Ministry of Higher Education and ASRT are the principal institutional funding pathways for follow-on development phases, including flight model fabrication and launch.

8.3 Research Centers & Organizations

At the national research level, the Egyptian Space Agency (EgSA) is the single most strategically significant stakeholder. EGOS-1 constitutes a direct contribution to EgSA's mandate to develop indigenous satellite capabilities, and the testbed provides a replicable testing infrastructure that could serve future national microsatellite programs. NARSS and NRIAG are the primary scientific end-users of the gamma-ray observation data that EGOS-1 would produce in orbit. The National Research Centre and the Space Science Center at Cairo University represent further collaboration and knowledge-transfer opportunities within Egypt's established research ecosystem.

9. Conclusion

9.1 EGOS-1 Micro-Satellite

The EGOS-1 micro-satellite project successfully demonstrated the feasibility of designing and fabricating a scientifically capable, LEO-compatible space observatory platform within an undergraduate academic setting. The 20 × 20 × 20 cm Aluminum 6061-T6 structural frame was fabricated via CNC laser cutting and validated through finite-element analysis, confirming a factor of safety of 7 under 10G launch loading, a peak thermal exposure of 411.51 K within material limits, and natural frequencies sufficiently separated from typical launch vehicle excitation bands.

The full electronic architecture — encompassing the ESP32 OBC, MPU-6050 IMU, HMC5883L magnetometer, reaction wheel, and dual magnetorquers — was assembled and validated at the subsystem level. A complementary filter for attitude determination and a PSO-tuned PD controller for stabilization were implemented and tested, with real-time telemetry verified through the onboard dashboard. The NaI(Tl) gamma-ray payload was completed at the design and simulation level; physical crystal fabrication was deferred due to manufacturing constraints outside the project's scope.

EGOS-1 constitutes a complete systems engineering case study spanning mission definition, orbit selection, subsystem design, simulation, fabrication, and ground testing, representing a tangible contribution to Egypt's developing indigenous satellite engineering capability.

9.2 AOCS Evaluation Testbed

The AOCS Evaluation Testbed met its primary objective of providing a functional, low-cost laboratory facility for validating satellite attitude control systems under LEO-representative conditions, and has been successfully commissioned at New Mansoura University.

The Helmholtz cage, wound to the optimal $\gamma = 0.5445$ coil separation ratio, generates field strengths within the LEO geomagnetic range (0.3–0.6 Gauss at 5 A) across all three orthogonal axes, with a uniform field region of ±0.45 m providing adequate clearance around the EGOS-1 mock-up. At 10 A, the cage reaches 1 Gauss, enabling over-range calibration and sensor saturation testing. The spherical air bearing was validated by two independent CFD analyses in agreement to within 0.01%, establishing a clear simulation baseline. Both subsystems were successfully fabricated from domestically sourced materials, demonstrating that a fully functional AOCS test facility can be realized at costs accessible to university programs and replicated at other Egyptian research institutions.

10.Future Work

Based on the limitations identified during the current project phase, the following areas are prioritized for future development:

Payload Fabrication and Integration. Physical fabrication of the NaI(Tl) scintillation crystal was not completed due to manufacturing constraints. Future work should procure a radiation-hardened crystal, complete payload integration, and perform end-to-end characterization including energy resolution calibration and detection efficiency measurement.

Solar Panel Integration. Dimensional mismatch between delivered panels and design specifications prevented solar panel integration. Future work should source conforming panels or revise mounting interfaces, and validate full EPS operation under simulated solar illumination.

Structural Material Upgrade. The prototype frame was fabricated in Stainless Steel 304 as a substitute for Al 6061-T6 due to material availability. Future work should repeat fabrication in the correct flight-grade alloy and conduct physical vibration testing to experimentally validate the FEA results.

Air Bearing Lifting Force Enhancement. The current 13 × 1.0 mm orifice configuration achieves 49.85 N against the 147.2 N design target, limited by choked-flow conditions. Future work should evaluate enlarged orifice diameters (1.25–1.50 mm) and increased orifice count to raise lifting capacity to the full operational requirement.

Closed-Loop AOCS Validation. Component-level testbed validation should be extended to full closed-loop testing, with the satellite mock-up mounted on the air bearing inside the Helmholtz cage, to experimentally verify controller performance under realistic simulated orbital magnetic conditions.

Advanced Control Algorithms. Future iterations should investigate model predictive control (MPC) or reinforcement learning-based architectures to improve disturbance rejection, reduce steady-state pointing error, and manage reaction wheel momentum saturation more effectively than the current PD implementation.

Onboard AI for Science Operations. A lightweight onboard inference model should be deployed to perform real-time gamma-ray event classification and automated downlink prioritization, reducing raw data volume within the constrained communication budget.

Flight Model Development and Launch Campaign. The long-term program objective is a flight-qualified EGOS-1 and deployment into the target 550 km sun-synchronous orbit. This requires a formal flight model development program — including thermal-vacuum, vibration, and EMC qualification testing — and early engagement with the Egyptian Space Agency and launch service providers to identify a suitable launch opportunity.

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